

MIMO Aircraft Longitudinal Control and Altitude Tracking Autopilot Design

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Abstract—Longitudinal flight control is central to aircraft stability and autonomous altitude regulation, yet classical cascaded PID autopilots require extensive manual tuning and scale poorly to coupled multi-input multi-output (MIMO) dynamics. This work develops a complete longitudinal autopilot for a fixed-wing aircraft using a Linear Quadratic Gaussian (LQG) control architecture. A five-state longitudinal state-space model is first constructed from the published stability and control derivatives of NASA Flight Condition 7 (Heffley and Jewell). An optimal Linear Quadratic Regulator (LQR) is designed to improve the damping of the short-period and phugoid modes, and a discrete-time-equivalent Kalman filter is developed to reconstruct the full state vector from noisy measurements of forward velocity and altitude alone. The LQR gain and Kalman observer are combined through the Separation Principle to form the inner-loop LQG controller, which is then embedded within a cascaded autopilot in which an outer-loop PI controller generates a saturation-limited pitch reference for altitude tracking. First-order actuator dynamics and control-surface saturation are further introduced to improve simulation realism. Closed-loop results show a maximum altitude overshoot of 0.235% for a 100 m altitude-hold command, bounded and smooth elevator/throttle commands, and estimation errors that remain small throughout both disturbance rejection and tracking manoeuvres. The results demonstrate that a modern state-space control architecture can deliver accurate, disturbance-robust altitude-hold performance within the validity region of the linearized aircraft model.

Index Terms—Linear Quadratic Regulator, Kalman Filter, Linear Quadratic Gaussian Control, Aircraft Longitudinal Dynamics, Altitude Hold Autopilot, State Estimation

I. INTRODUCTION AND MOTIVATION

Modern aircraft rely extensively on automatic flight control systems to enhance stability, reduce pilot workload, and improve flight safety. Among the fundamental control tasks performed by an autopilot, longitudinal flight control plays a central role by regulating the aircraft's pitch attitude, flight path, and altitude. Accurate longitudinal control enables aircraft to maintain desired flight conditions despite external disturbances, sensor noise, and modelling uncertainties.

Traditional aircraft autopilots have historically employed cascaded proportional-integral-derivative (PID) controllers due to their simplicity and ease of implementation. While effective for many applications, classical controllers often require extensive manual tuning and may struggle to provide optimal performance for coupled multi-variable systems. Modern control techniques based on state-space modelling offer a systematic framework for handling multiple interacting states while explicitly accounting for system dynamics.

Linear Quadratic Gaussian (LQG) control combines two powerful concepts: the Linear Quadratic Regulator (LQR),

which computes an optimal state-feedback control law by minimizing a quadratic performance index, and the Kalman Filter, which provides optimal state estimation in the presence of process and measurement noise. Together, these techniques form a robust output-feedback controller capable of achieving excellent tracking performance even when all system states are not directly measurable.

The objective of this project is to design and implement a complete longitudinal autopilot for a fixed-wing aircraft using an LQG control architecture. Beginning from a linearized aircraft model derived from published flight dynamic data, the project develops an optimal state-feedback controller, designs a Kalman observer for state estimation, and integrates both into a complete output-feedback control system. An outer-loop altitude controller is subsequently introduced to enable autonomous altitude tracking while maintaining satisfactory pitch dynamics.

To improve the practical relevance of the controller, actuator dynamics, control surface saturation, process disturbances, and sensor noise are incorporated into the simulation environment. The resulting autopilot is evaluated through extensive simulation studies, including disturbance rejection, state estimation accuracy, control effort analysis, and altitude tracking performance.

A. Project Contributions

The primary contributions of this work are summarized as follows:

- Development of a five-state longitudinal aircraft model from published flight dynamic data.
- Design of an optimal Linear Quadratic Regulator (LQR) for longitudinal state feedback control.
- Implementation of a discrete Kalman Filter for real-time estimation of non-measurable aircraft states.
- Integration of the LQR controller and Kalman observer to form a complete Linear Quadratic Gaussian (LQG) controller.
- Design of a cascaded altitude hold autopilot using an outer-loop pitch command generator.
- Inclusion of actuator dynamics, actuator saturation, process disturbances, and measurement noise to improve simulation realism.
- Comprehensive performance evaluation using time-domain response, pole analysis, state estimation accuracy, and control effort metrics.

B. Overall Autopilot Architecture

Figure 1 presents the overall control architecture developed in this project. The autopilot employs a cascaded control structure in which an outer-loop altitude controller generates the desired pitch command, while an inner-loop Linear Quadratic Gaussian (LQG) controller tracks the commanded pitch angle. State estimation is performed using a Kalman filter, and first-order actuator dynamics are included to model realistic control surface behaviour.

II. PROJECT SETUP AND MODELLING

The first stage of the project involved developing a mathematical model of the aircraft longitudinal dynamics. Since modern control techniques such as the Linear Quadratic Regulator (LQR) and Kalman filtering require a state-space representation of the plant, an accurate linear model of the aircraft was obtained from published flight dynamic data.

The aircraft model employed in this work is based on the NASA Heffley and Jewell aircraft dynamics report, which provides linearized stability and control derivatives for several flight conditions. The published dimensional stability derivatives were used to construct the complete longitudinal state-space model, forming the basis for all subsequent controller and observer design.

Beginning from the linear aircraft model derived in this section, successive stages of the project introduce optimal state feedback using an LQR controller, state estimation through a Kalman filter, and finally an outer-loop altitude hold controller with actuator dynamics and command saturation. The mathematical model developed here therefore serves as the foundation upon which the entire control system is built.

A. Aircraft Model

Aircraft motion is commonly separated into longitudinal and lateral-directional dynamics. Since the objective of this work is the design of an altitude hold autopilot, only the longitudinal dynamics are considered. This assumption significantly reduces the model complexity while retaining the dominant dynamics governing pitch motion and altitude control.

The longitudinal state vector is defined as

$$x = [u \quad w \quad q \quad \theta \quad \Delta h]^T \quad (1)$$

where

- u : forward velocity perturbation (m/s),
- w : vertical velocity perturbation (m/s),
- q : pitch rate (rad/s),
- θ : pitch angle (rad),
- Δh : change in altitude (m).

Unlike many aircraft dynamics references that directly provide the state-space matrices, the NASA report presents the aircraft equations in descriptor form together with the dimensional stability derivatives. The state-space representation used throughout this work was therefore constructed directly from the published derivatives.

B. Flight Condition

The aircraft data correspond to **Flight Condition 7** of the NASA Heffley and Jewell report (Table IX-4). The trim condition represents steady, wings-level flight under the following operating conditions.

TABLE I: Flight Condition 7

Parameter	Value
Altitude	20,000 ft
Mach Number	0.80
Trim Forward Velocity U_0	820.01 ft/s
Trim Vertical Velocity W_0	0 ft/s
Trim Pitch Angle θ_0	0 rad
Gravitational Acceleration g	32.174 ft/s ²

C. Dimensional Stability Derivatives

The longitudinal dimensional stability derivatives obtained from the NASA report are listed in Table II. These derivatives characterize the sensitivity of aerodynamic forces and pitching moment to perturbations in the aircraft states.

Similarly, the elevator and throttle control derivatives are listed in Table III.

TABLE III: Control Derivatives

Derivative	Elevator	Throttle
X	0	5.05×10^{-4}
Z	-32.7	-2.20×10^{-6}
M	-2.03	3.02×10^{-6}

D. Governing Differential Equations

The dimensional stability and control derivatives listed in the previous section describe the linearized aerodynamic force and moment characteristics of the aircraft about the selected trim condition. Using these derivatives, the longitudinal dynamics may be expressed by the following set of coupled linear differential equations.

The forward velocity perturbation is governed by

$$\Delta \dot{u} = X_u \Delta u + X_w \Delta w - g \cos \theta_0 \Delta \theta + X_{\delta_e} \delta_e + X_{\delta_T} \delta_T, \quad (2)$$

where X_u and X_w represent the aerodynamic force derivatives with respect to forward and vertical velocity perturbations, respectively, while the final two terms account for the effects of elevator and throttle inputs.

The vertical velocity dynamics are described by

$$(1 - Z_{\dot{w}}) \Delta \dot{w} = Z_u \Delta u + Z_w \Delta w + (U_0 + Z_q) \Delta q - g \sin \theta_0 \Delta \theta + Z_{\delta_e} \delta_e + Z_{\delta_T} \delta_T \quad (3)$$

where the acceleration derivative $Z_{\dot{w}}$ introduces the descriptor formulation adopted by the NASA report.

Similarly, the pitching moment equation is

$$\Delta \dot{q} = M_u \Delta u + M_w \Delta w + M_q \Delta q + M_{\dot{w}} \Delta \dot{w} + M_{\delta_e} \delta_e + M_{\delta_T} \delta_T, \quad (4)$$

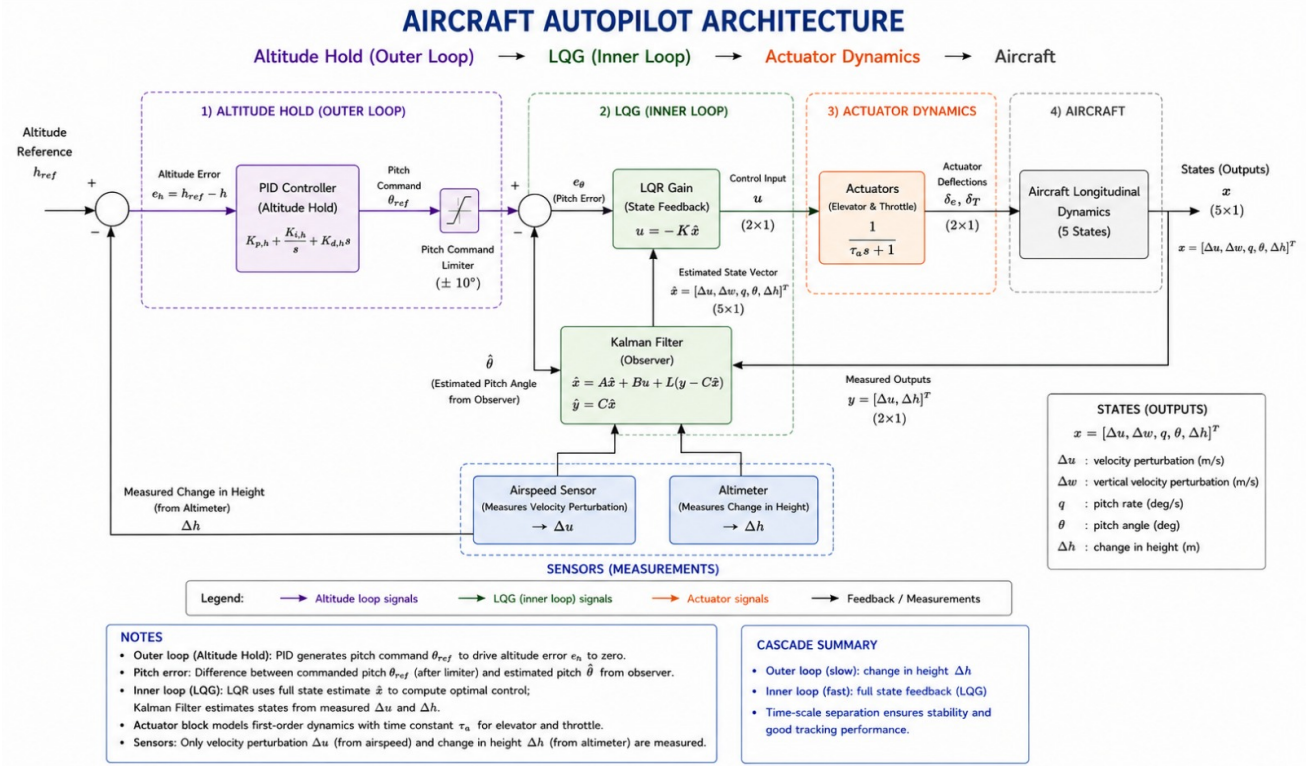


Fig. 1.1. Overall architecture of the proposed cascaded longitudinal autopilot.

TABLE II: Longitudinal Dimensional Stability Derivatives

X_u	Z_u	M_u	X_w	Z_w	M_w	$Z_{\dot{w}}$	$M_{\dot{w}}$	X_q	Z_q	M_q
-0.00643	-0.0941	-0.000222	0.0253	-0.624	-0.00153	0.0144	-0.000212	0	-9.99	-0.658

where $M_{\dot{w}}$ represents the pitching moment contribution arising from vertical acceleration.

The pitch kinematics are obtained from

$$\Delta \dot{\theta} = \Delta q. \quad (5)$$

Finally, altitude is introduced through the longitudinal kinematic relationship

$$\Delta \dot{h} = U_0 \Delta \theta - \Delta w, \quad (6)$$

which augments the original four-state aircraft model to produce the five-state longitudinal representation employed throughout this work.

Equations (2)–(6) completely describe the linearized longitudinal dynamics about the selected trim condition. Since the acceleration derivatives $Z_{\dot{w}}$ and $M_{\dot{w}}$ appear explicitly in the equations, the system is naturally expressed in descriptor form. In the following subsection, these equations are assembled into the descriptor matrix representation reported by NASA, which is subsequently converted into the standard state-space form used for controller and observer design.

E. Linearized State-Space Representation

The governing equations presented in the previous subsection may be written compactly in descriptor form by collecting the acceleration derivatives into the descriptor matrix. Defining the longitudinal state vector as

$$x = [\Delta u \quad \Delta w \quad \Delta q \quad \Delta \theta]^T, \quad (7)$$

and the control input vector as

$$u = [\delta_e \quad \delta_T]^T, \quad (8)$$

the aircraft dynamics can be expressed as

$$E \dot{x} = A'x + B'u, \quad (9)$$

where

$$E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 - Z_{\dot{w}} & 0 & 0 \\ 0 & -M_{\dot{w}} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (10)$$

$$A' = \begin{bmatrix} X_u & X_w & 0 & -g \cos \theta_0 \\ Z_u & Z_w & U_0 + Z_q & -g \sin \theta_0 \\ M_u & M_w & M_q & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad (11)$$

and

$$B' = \begin{bmatrix} X_{\delta_e} & X_{\delta_T} \\ Z_{\delta_e} & Z_{\delta_T} \\ M_{\delta_e} & M_{\delta_T} \\ 0 & 0 \end{bmatrix}. \quad (12)$$

For Flight Condition 7, the numerical descriptor matrices obtained from the NASA Heffley report are

$$E = \begin{bmatrix} 1.0000 & 0 & 0 & 0 \\ 0 & 0.9856 & 0 & 0 \\ 0 & 0.0002 & 1.0000 & 0 \\ 0 & 0 & 0 & 1.0000 \end{bmatrix}, \quad (13)$$

$$A' = \begin{bmatrix} -0.0064 & 0.0253 & 0 & -32.1740 \\ -0.0941 & -0.6240 & 810.0200 & 0 \\ -0.0002 & -0.0015 & -0.6580 & 0 \\ 0 & 0 & 1.0000 & 0 \end{bmatrix}, \quad (14)$$

and

$$B' = \begin{bmatrix} 0 & 0.0005 \\ -32.7000 & 0 \\ -2.0300 & 0.0000 \\ 0 & 0 \end{bmatrix}. \quad (15)$$

The descriptor representation is converted into the standard state-space form by premultiplying both sides of (9) by the inverse of the descriptor matrix,

$$\dot{x} = Ax + Bu, \quad (16)$$

where

$$A = E^{-1}A', \quad B = E^{-1}B'. \quad (17)$$

The published aircraft data are expressed entirely in Imperial units. Consequently, all dimensional quantities were converted into SI units before controller synthesis. The forward and vertical velocity states were converted from feet per second to metres per second, gravitational acceleration was replaced by its SI equivalent, and all the dimensional derivatives and trim quantities were converted consistently to SI units using standard conversion factors before constructing the state-space matrices.

The original NASA model describes only the four longitudinal states Δu , Δw , Δq , and $\Delta \theta$. Since the objective of this work is altitude hold, the model was augmented with an additional altitude state using the kinematic relationship

$$\Delta \dot{h} = U_0 \Delta \theta - \Delta w, \quad (18)$$

resulting in a complete five-state longitudinal model.

Finally, the throttle input provided in the NASA database corresponds to an engine thrust perturbation expressed in pounds-force (lbf). For controller implementation, this input was normalized with respect to the trim engine thrust so that the second control input represents a fractional throttle perturbation rather than an absolute thrust change. This normalization simplifies controller tuning and results in a dimensionless throttle command.

The resulting five-state system matrices used throughout the remainder of this work are

$$A = \begin{bmatrix} -0.0064 & 0.0253 & 0 & -9.8066 & 0 \\ -0.0955 & -0.6331 & 250.5013 & 0 & 0 \\ -0.0007 & -0.0046 & -0.8322 & 0 & 0 \\ 0 & 0 & 1.0000 & 0 & 0 \\ 0 & -1.0000 & 0 & 249.9400 & 0 \end{bmatrix}, \quad (19)$$

and

$$B = \begin{bmatrix} 0 & 0.8438 \\ -10.1126 & -0.0037 \\ -2.0230 & 0.0543 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}. \quad (20)$$

The resulting five-state, two-input state-space model forms the plant used throughout the remainder of this work for controller synthesis, state estimation, and altitude hold autopilot development.

III. OPEN-LOOP DYNAMIC ANALYSIS

Before designing the controller, the dynamic characteristics of the aircraft were investigated in open loop. This provides an understanding of the inherent longitudinal behaviour of the aircraft and establishes a baseline against which the performance of the proposed controller can be evaluated. Two disturbance scenarios were considered: an initial pitch angle disturbance and an initial pitch rate disturbance. In both cases, no control input was applied, allowing the natural aircraft dynamics to govern the response.

A. Response to an Initial Pitch Angle Disturbance

The first simulation considers an initial pitch angle perturbation of 5° while all remaining states are initialized to zero. This represents a disturbance that momentarily changes the aircraft attitude from its trim condition without any pilot or autopilot intervention.

Figure 2 illustrates the resulting evolution of the longitudinal states. The initial pitch disturbance produces noticeable coupling between the translational and rotational dynamics, causing perturbations in forward velocity, vertical velocity and pitch rate. Although all states eventually decay towards their equilibrium values, the responses exhibit lightly damped oscillatory behaviour with long settling times. The altitude response is particularly significant, demonstrating that even a relatively small pitch disturbance results in substantial deviation from the desired flight path.

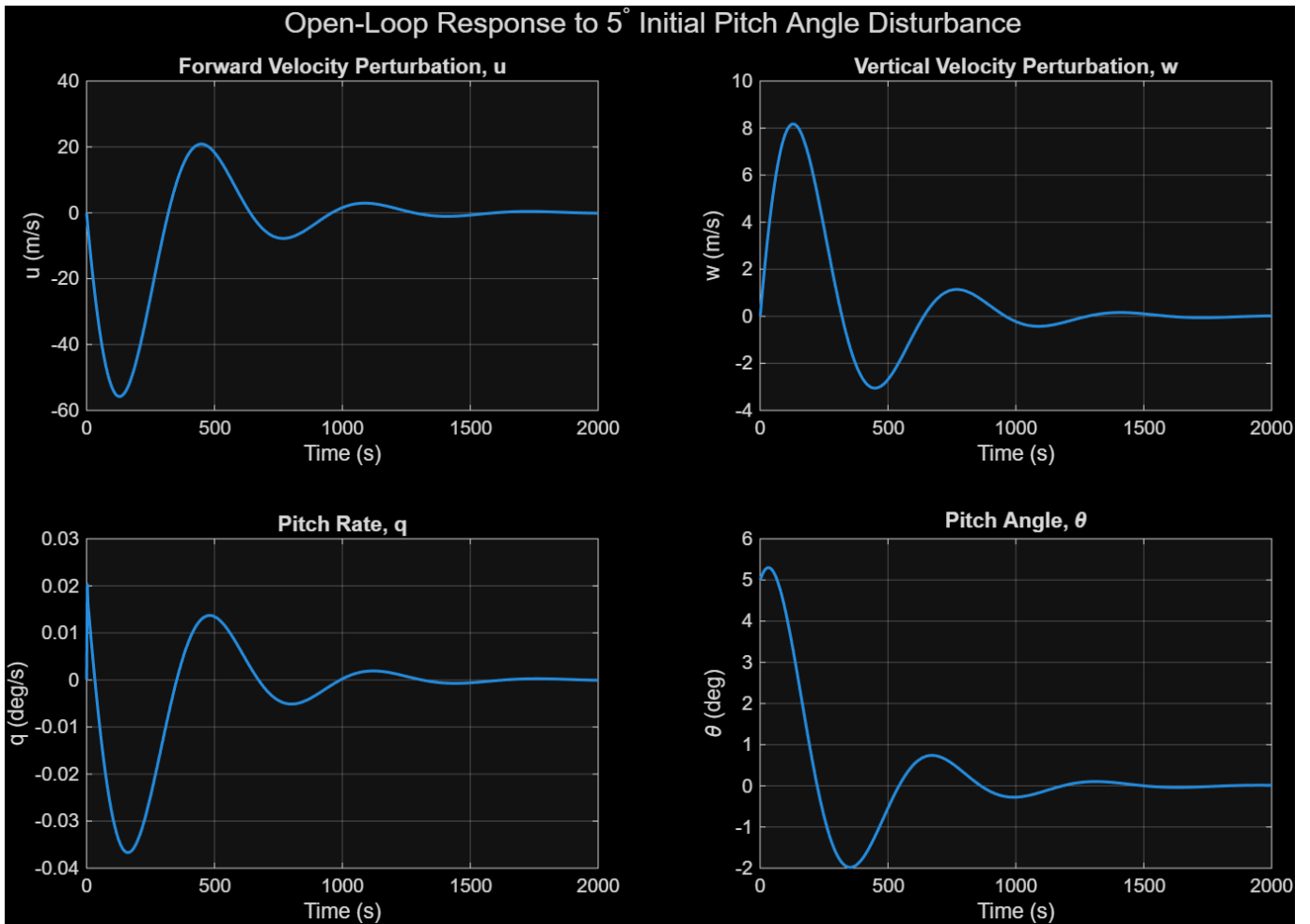


Fig. 3.1. Open-loop response of the aircraft to a 5° initial pitch angle disturbance.

The slow decay of the oscillations indicates that, while the aircraft is open-loop stable, its natural dynamic response is insufficient for autonomous flight control applications where rapid disturbance rejection and accurate trajectory tracking are required.

B. Response to an Initial Pitch Rate Disturbance

The second disturbance scenario considers an initial pitch rate of $5^\circ/\text{s}$ with all remaining states initialized at their trim values.

As shown in Fig. 3, the initial rotational disturbance excites both the short-period and phugoid modes of the aircraft. Immediately following the disturbance, a rapid transient is observed in the pitch rate response. This short-lived spike decays almost instantaneously and corresponds to the aircraft's highly damped short-period mode, which dominates the initial rotational dynamics. Following the decay of this fast transient, the response is governed by the much slower phugoid mode, resulting in low-frequency oscillations in the velocity perturbations, pitch attitude, and altitude.

Although the aircraft eventually returns to its equilibrium condition, the transient response exhibits slow damping and

significant altitude excursions, indicating poor disturbance rejection capability in open loop. These results demonstrate that, despite possessing inherent dynamic stability, the natural aircraft dynamics are inadequate for precision flight control applications where rapid recovery from disturbances and accurate trajectory regulation are required.

C. Altitude Deviation Following Disturbances

Although the complete state responses provide insight into the dynamic behaviour of the aircraft, the quantity of primary interest for an autopilot is the resulting altitude deviation. Even relatively small attitude disturbances can accumulate into substantial altitude errors, making altitude an important performance indicator for longitudinal flight control.

Figure 4 shows the altitude response following an initial pitch angle disturbance, while Fig. 5 presents the corresponding response for an initial pitch rate disturbance.

It is evident that both disturbances produce significant altitude excursions before the aircraft slowly returns towards its equilibrium flight condition. While the aircraft remains dynamically stable, the large altitude deviations and long settling times demonstrate that the natural dynamics alone

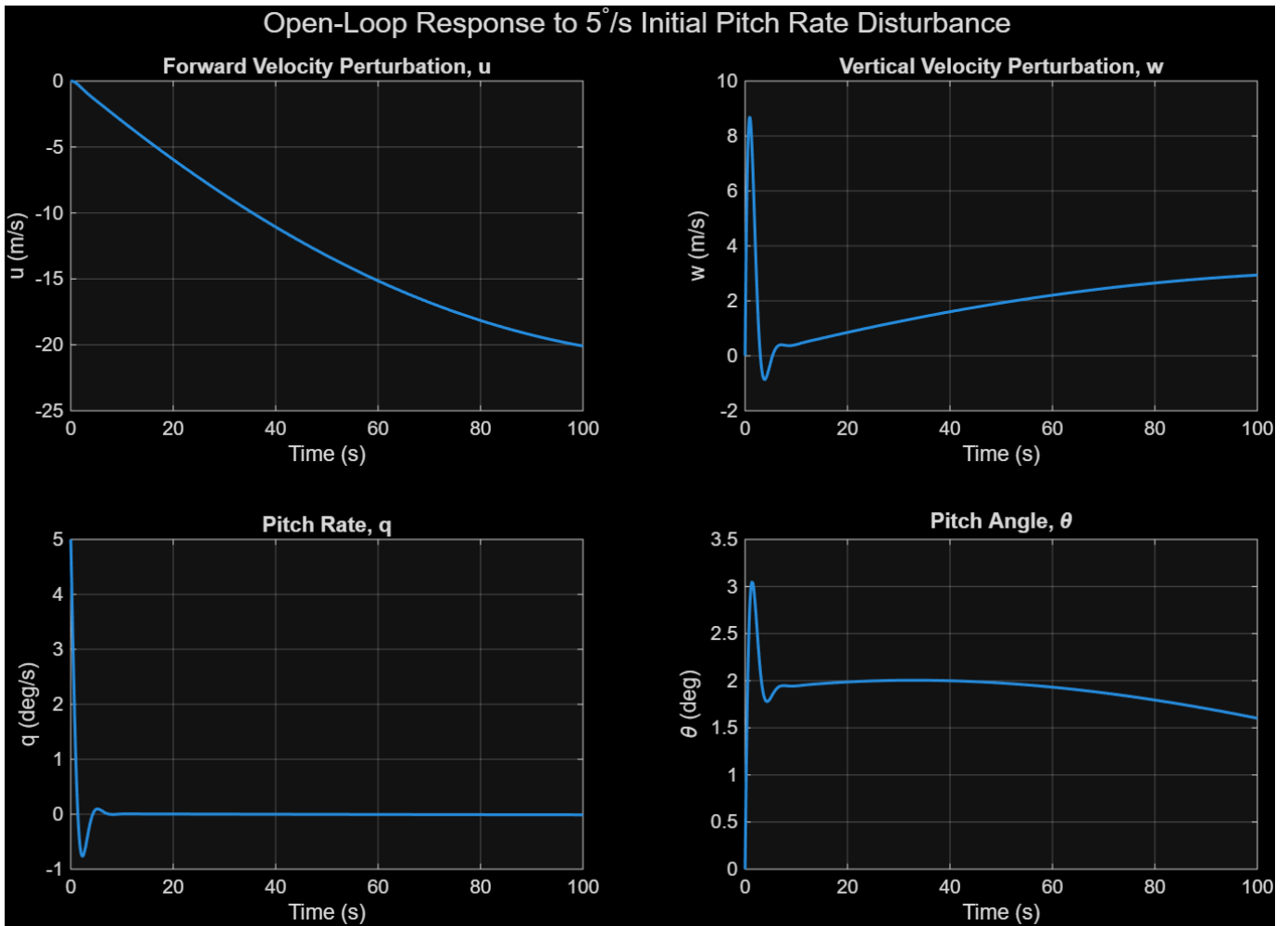


Fig. 3.2. Open-loop response of the aircraft to a $5^\circ/\text{s}$ initial pitch rate disturbance.

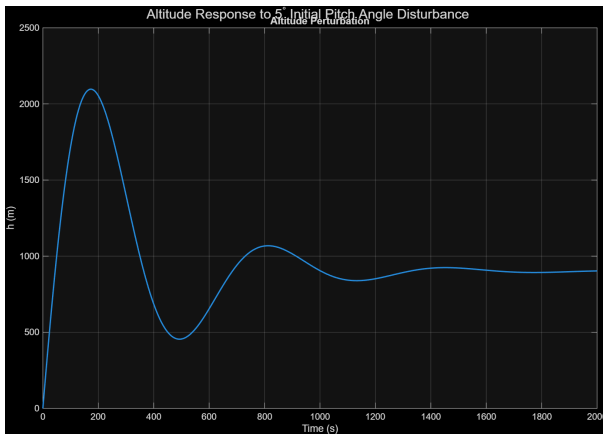


Fig. 3.3. Open-loop altitude response to an initial 5° pitch angle disturbance.

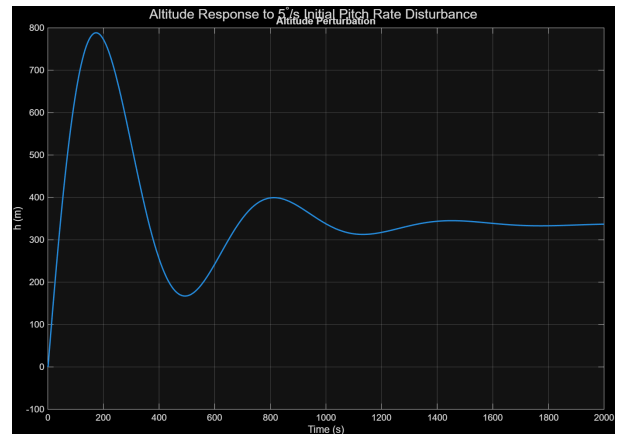


Fig. 3.4. Open-loop altitude response to an initial $5^\circ/\text{s}$ pitch rate disturbance.

are inadequate for precision altitude regulation. These observations further motivate the development of an active feed-back controller capable of rapidly rejecting disturbances while

maintaining the desired flight path.

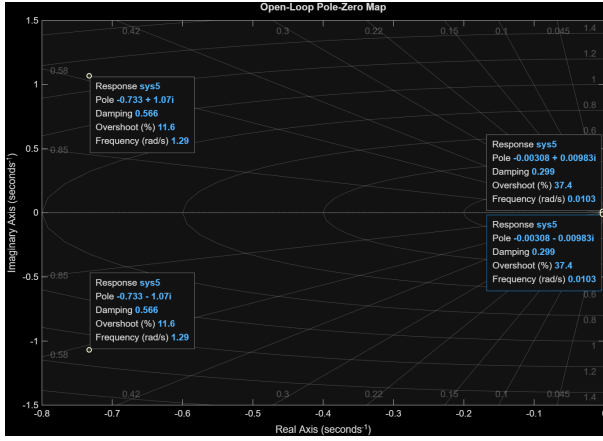


Fig. 3.5. Open-loop pole-zero map of the longitudinal aircraft model.

TABLE IV: Open-loop longitudinal modes of the aircraft

Mode	Eigenvalue	ζ	ω_n (rad/s)
Altitude Integrator	0	–	0.0000
Short-Period	$-0.7328 \pm 1.0676i$	0.5659	1.2949
Phugoid	$-0.0031 \pm 0.0098i$	0.2990	0.0103

D. Open-Loop Pole Analysis

To further investigate the dynamic characteristics of the aircraft, the eigenvalues of the open-loop system were examined using the pole-zero map shown in Fig. 6.

The pole locations confirm the observations obtained from the time-domain responses. All poles lie within the left-half of the complex plane, indicating that the aircraft is dynamically stable about the selected trim condition. However, several poles are located close to the imaginary axis, resulting in lightly damped oscillatory modes and slow transient decay. These characteristics are reflected in the long settling times observed in Figs. 2 and 3.

The dominant dynamic modes were further characterized by evaluating their natural frequencies and damping ratios. The corresponding values are summarized in Table IV.

The eigenvalue analysis confirms the observations obtained from the time-domain simulations. The short-period mode possesses a relatively high natural frequency and moderate damping ratio, causing the rapid oscillations observed immediately following a disturbance. In contrast, the phugoid mode exhibits a very low natural frequency and comparatively small damping ratio, resulting in the slow oscillatory behaviour and large altitude excursions evident in the open-loop responses.

The additional pole at the origin arises from the augmentation of the longitudinal model with the altitude state. Since altitude is obtained by integrating the vertical flight path, it naturally introduces an integrator into the state-space representation without affecting the dynamic stability of the aircraft.

IV. LINEAR QUADRATIC REGULATOR DESIGN

A. Design Objective

The open-loop analysis presented in the previous section demonstrated that although the aircraft is dynamically stable about its trim condition, its natural response is characterized by lightly damped oscillations and long settling times. These characteristics are primarily associated with the aircraft's short-period and phugoid modes, each of which governs a distinct aspect of the longitudinal dynamics.

The short-period mode is a relatively fast oscillatory mode that dominates the aircraft's immediate response following a pitch disturbance. As observed in the open-loop pitch rate response, the aircraft experiences a rapid transient immediately after the disturbance before the motion transitions to slower dynamics. While this mode is reasonably damped, inadequate damping can result in rapid attitude oscillations, degraded handling qualities, and increased pilot workload, particularly during aggressive maneuvers or atmospheric disturbances.

In contrast, the phugoid mode represents a slow exchange between the aircraft's kinetic and potential energy. This mode possesses a very low natural frequency and comparatively small damping ratio, causing the aircraft to exhibit long-period oscillations in forward velocity, altitude and pitch attitude. Although the oscillations eventually decay, the prolonged settling time results in poor disturbance rejection and large altitude deviations, making the open-loop aircraft unsuitable for precision flight control tasks such as altitude hold.

These observations motivate the use of active feedback control to improve the transient response of the aircraft. The controller should simultaneously

- increase the damping of the dominant longitudinal modes,
- reduce settling time following disturbances,
- minimize oscillatory behaviour,
- regulate all longitudinal states about their trim values, and
- achieve these objectives without demanding excessive control effort from the elevator or throttle actuators.

Since the longitudinal aircraft dynamics constitute a coupled multi-input multi-output (MIMO) system, classical single-loop tuning methods become increasingly difficult to apply while simultaneously achieving satisfactory performance across all states. An optimal state-feedback controller based on the Linear Quadratic Regulator (LQR) was therefore selected. Unlike classical pole-placement techniques, the LQR framework provides a systematic means of balancing state regulation against actuator effort through the minimization of a quadratic cost function. This produces a feedback controller that improves the dynamic performance of the aircraft while accounting for the interactions between the longitudinal states.

B. LQR Formulation

The Linear Quadratic Regulator (LQR) is an optimal state-feedback control technique that determines the control law by minimizing a quadratic performance index. Unlike classical controller design methods, which typically rely on manual pole placement or frequency-domain analysis, the LQR framework

computes an optimal feedback gain by simultaneously considering state regulation and control effort.

The longitudinal aircraft dynamics are represented in the standard state-space form

$$\dot{x} = Ax + Bu, \quad (21)$$

where $x \in \mathbb{R}^5$ denotes the state vector and $u \in \mathbb{R}^2$ represents the control input vector consisting of elevator deflection and throttle command.

The optimal control law is chosen as

$$u = -Kx, \quad (22)$$

where K is the optimal feedback gain matrix obtained by minimizing the quadratic cost function

$$J = \int_0^\infty (x^T Q x + u^T R u) dt. \quad (23)$$

The weighting matrices Q and R determine the relative importance assigned to state regulation and control effort. Large entries in Q force the controller to reduce deviations in the corresponding states more aggressively, while larger values in R discourage excessive actuator usage by increasing the penalty associated with the control inputs. Consequently, the controller design becomes a trade-off between transient performance and actuator effort.

The optimal state-feedback gain matrix is obtained by solving the continuous-time Algebraic Riccati Equation (ARE),

$$A^T P + PA - PBR^{-1}B^T P + Q = 0, \quad (24)$$

where P is the unique positive-definite solution of the Riccati equation. The corresponding optimal feedback gain is then computed as

$$K = R^{-1}B^T P. \quad (25)$$

C. Selection of Weighting Matrices

The selection of suitable weighting matrices represents one of the most important aspects of LQR controller design. As an initial estimate, the state and input weighting matrices were selected using *Bryson's Rule*, which provides a systematic method for normalizing variables with different physical units.

Bryson's Rule assigns the diagonal entries of the weighting matrices as

$$Q_{ii} = \frac{1}{x_{i,\max}^2}, \quad R_{jj} = \frac{1}{u_{j,\max}^2}, \quad (26)$$

where $x_{i,\max}$ denotes the maximum acceptable deviation of the i^{th} state and $u_{j,\max}$ represents the maximum allowable control input.

Substituting (26) into the quadratic cost function gives

$$x_i^2 Q_{ii} = \frac{x_i^2}{x_{i,\max}^2}. \quad (27)$$

Therefore, when a state reaches its prescribed maximum allowable deviation,

$$x_i = x_{i,\max}, \quad (28)$$

its contribution to the performance index becomes

$$x_i^2 Q_{ii} = \frac{x_{i,\max}^2}{x_{i,\max}^2} = 1. \quad (29)$$

Similarly,

$$u_j^2 R_{jj} = \frac{u_j^2}{u_{j,\max}^2}, \quad (30)$$

so that each control input also contributes unity to the cost function when operating at its maximum acceptable value. This normalization ensures that every state and control input is weighted relative to its permissible operating range rather than its numerical magnitude or physical units.

The weighting matrices obtained using Bryson's Rule served as an initial design point. However, because the longitudinal aircraft states exhibit significantly different dynamic characteristics and the actuator authority of the elevator and throttle differs considerably, manual tuning of the weighting matrices was subsequently performed. The final weighting matrices, obtained through iterative simulation to achieve a satisfactory compromise between disturbance rejection, transient response and control effort, are

$$Q = \begin{bmatrix} 0.0100 & 0 & 0 & 0 & 0 \\ 0 & 0.0400 & 0 & 0 & 0 \\ 0 & 0 & 32.8281 & 0 & 0 \\ 0 & 0 & 0 & 131.3123 & 0 \\ 0 & 0 & 0 & 0 & 0.0100 \end{bmatrix}, \quad (31)$$

and

$$R = \begin{bmatrix} 820.7016 & 0 \\ 0 & 100 \end{bmatrix}. \quad (32)$$

Greater weights were assigned to the pitch rate and pitch angle states to suppress oscillations and improve the aircraft's attitude response, while comparatively smaller weights were assigned to the translational velocity and altitude states. Since altitude regulation is performed by the outer-loop controller, the inner-loop LQR was designed primarily for rapid attitude stabilization. The control weighting matrix was selected to limit excessive elevator and throttle activity while maintaining adequate control authority. Note that the larger elevator weight $R_{11} = 820.70$ relative to the throttle weight $R_{22} = 100$ does not indicate a stronger penalty on elevator effort in an absolute sense; rather, it follows directly from Bryson's Rule, since the elevator's acceptable deflection range (in radians) is numerically much smaller than the throttle's acceptable fractional range, so $1/u_{\max}^2$ is correspondingly larger for the elevator channel.

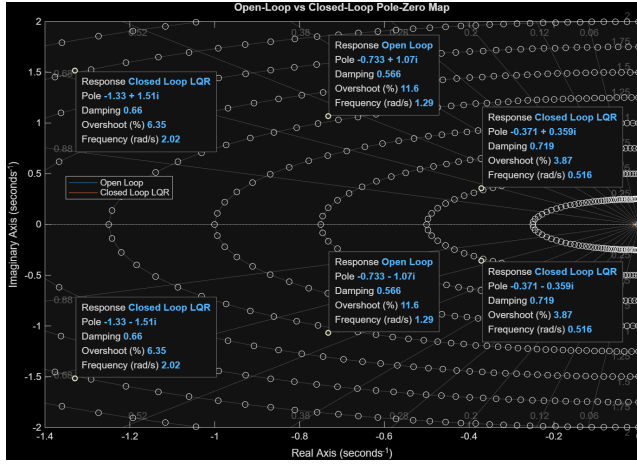


Fig. 4.1. Comparison of open-loop and LQR closed-loop pole locations.

TABLE V: Closed-loop eigenvalue characteristics

Mode	Eigenvalue	ζ	ω_n (rad/s)
Short-Period	$-1.3292 \pm 1.5146i$	0.6596	2.0151
Slow Oscillatory Mode	$-0.3711 \pm 0.3586i$	0.7192	0.5161
Real Mode	-0.0099	1.0000	0.0099

D. Closed-Loop Pole Analysis

Following the synthesis of the LQR controller, the closed-loop dynamics were examined by analysing the eigenvalues of the resulting system. The application of optimal state feedback shifts the dominant poles further into the left-half of the complex plane, thereby increasing system damping and improving the transient response of the aircraft.

Figure 7 compares the open-loop and closed-loop pole locations. It is evident that the LQR controller relocates the dominant longitudinal modes to positions associated with faster decay and reduced oscillatory behaviour.

The eigenvalues of the closed-loop system are

$$\lambda(A_{cl}) = \begin{cases} -1.3292 \pm 1.5146i, \\ -0.3711 \pm 0.3586i, \\ -0.0099. \end{cases} \quad (33)$$

The corresponding modal characteristics are summarized in Table V.

Compared with the open-loop aircraft dynamics, the dominant poles have moved significantly further into the left-half plane, resulting in higher damping ratios and increased natural frequencies. The short-period mode now exhibits a damping ratio of approximately 0.66, allowing rapid attenuation of pitch disturbances while maintaining a well-behaved transient response.

It is also observed that the slower longitudinal dynamics have been preserved following controller synthesis. Rather than eliminating the phugoid behaviour entirely, the LQR controller relocates the associated poles to improve damping while retaining the characteristic slow longitudinal motion of the

aircraft. Preserving the phugoid is desirable since it represents the natural long-period exchange between kinetic and potential energy and provides a smooth response to changes in flight conditions. This also allows the subsequent outer-loop altitude controller to regulate altitude without unnecessarily altering the fundamental aircraft dynamics.

Overall, the pole relocation demonstrates that the LQR controller successfully improves the dynamic stability of the aircraft while maintaining the essential characteristics of the longitudinal dynamics. The resulting increase in damping is reflected directly in the closed-loop time responses presented in the following subsection.

E. Closed-Loop Time Response

The effectiveness of the LQR controller was evaluated by subjecting the closed-loop aircraft to the same disturbance scenarios considered during the open-loop analysis. This allows a direct comparison of the system response before and after the introduction of optimal state feedback.

Figure 8 illustrates the evolution of the longitudinal states following the application of the LQR controller. In comparison with the open-loop response, all states converge significantly faster to their equilibrium values with substantially reduced oscillations.

The rapid decay of the pitch rate and pitch angle responses indicates that the controller effectively suppresses the short-period dynamics while simultaneously improving the damping of the slower longitudinal motion. The forward and vertical velocity perturbations also return to their trim conditions considerably faster than in the open-loop case, demonstrating improved disturbance rejection throughout the coupled system.

The corresponding elevator and throttle commands generated by the LQR controller are shown in Fig. 9. Both control inputs remain smooth and bounded throughout the simulation, indicating that the desired improvement in dynamic response is achieved without demanding unrealistic actuator effort.

The control inputs exhibit relatively large initial corrections immediately following the disturbance, after which both the elevator and throttle commands rapidly converge towards their trim values. This behaviour is expected since the controller applies an aggressive initial correction to suppress the disturbance before reducing the control effort as the aircraft approaches equilibrium.

F. Discussion

The LQR controller significantly improves the dynamic performance of the aircraft compared with the open-loop system. Increased damping of the dominant longitudinal modes results in faster settling, reduced oscillations and improved disturbance rejection, while maintaining smooth and bounded elevator and throttle commands.

A limitation of the LQR approach is its assumption that the complete state vector is available for feedback. In practice, several longitudinal states cannot be measured directly and must instead be estimated from available sensor measurements. To address this, a Kalman filter is designed in the

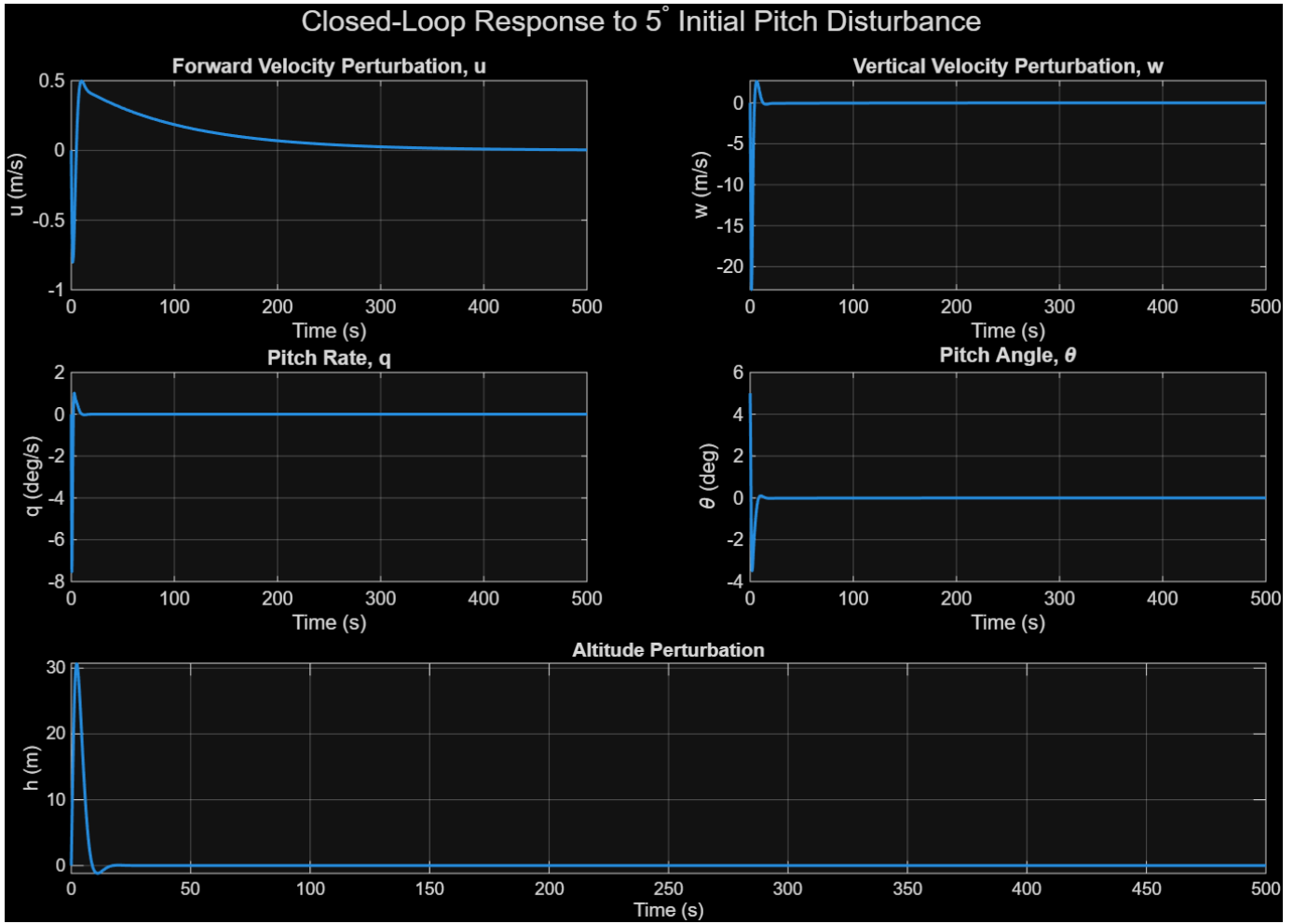


Fig. 4.2. Closed-loop longitudinal state response using the LQR controller.

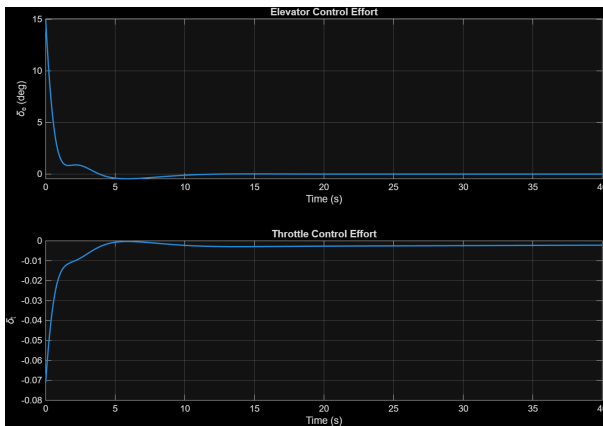


Fig. 4.3. Control effort generated by the LQR controller.

following section to provide optimal state estimates, which are subsequently combined with the LQR controller to form the complete Linear Quadratic Gaussian (LQG) controller.

V. KALMAN FILTER DESIGN

A. Motivation

The LQR controller developed in the previous section assumes that the complete aircraft state vector is available for feedback. In practice, however, this assumption is rarely satisfied. Modern aircraft are equipped with only a limited number of sensors, and several longitudinal states cannot be measured directly. Instead, these states must be estimated from the available measurements and a mathematical model of the aircraft.

Furthermore, sensor measurements are inevitably corrupted by noise, while the mathematical model itself cannot perfectly represent the true aircraft dynamics due to modelling uncertainties and external disturbances. Directly feeding noisy measurements to the controller can therefore degrade closed-loop performance and may even compromise system stability.

To address these challenges, a Kalman filter was designed to estimate the complete longitudinal state vector from the available measurements. The Kalman filter provides a statistically optimal state estimate by combining information from the mathematical model with noisy sensor measurements.

In the present work, measurements of the forward velocity perturbation and altitude were assumed to be available, while the remaining states were reconstructed by the observer.

The resulting state estimates are subsequently employed by the LQR controller, forming the complete Linear Quadratic Gaussian (LQG) controller described later in this report.

B. Process and Measurement Noise Modelling

The performance of a Kalman filter depends critically on the characterization of process disturbances and sensor measurement noise. Accordingly, stochastic noise sources were incorporated into the simulation to emulate practical flight conditions.

Process noise represents unmodelled aircraft dynamics and external disturbances such as atmospheric turbulence, wind gusts, modelling errors and aerodynamic uncertainties. These disturbances were modelled as zero-mean Gaussian white noise acting on the system dynamics,

$$\dot{x} = Ax + Bu + w, \quad (34)$$

where

$$w \sim \mathcal{N}(0, Q_k)$$

denotes the process noise vector with covariance matrix Q_k .

Similarly, sensor measurements were assumed to be corrupted by additive Gaussian white noise,

$$y = Cx + v, \quad (35)$$

where

$$v \sim \mathcal{N}(0, R_k)$$

is the measurement noise with covariance matrix R_k .

For this work, only the forward velocity perturbation and altitude were assumed to be directly measurable. These measurements were corrupted by independent Gaussian noise prior to being supplied to the Kalman filter, while the remaining states were reconstructed solely through observer dynamics.

C. Kalman Filter Formulation

The Kalman filter is an optimal state estimator that combines a mathematical model of the aircraft with noisy sensor measurements to produce minimum-variance estimates of the system states. Unlike direct measurement, which is susceptible to sensor noise, the Kalman filter continuously predicts the system evolution while correcting its estimate using available measurements.

The longitudinal aircraft dynamics are represented as

$$\dot{x} = Ax + Bu + Gw, \quad (36)$$

where w denotes the process disturbance vector and G is the process disturbance distribution matrix. The corresponding measurement equation is

$$y = Cx + v, \quad (37)$$

where v represents the measurement noise vector.

The observer estimates the aircraft states according to

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x}), \quad (38)$$

where $\hat{x}$ is the estimated state vector and L is the Kalman observer gain matrix.

The term

$$y - C\hat{x} \quad (39)$$

is referred to as the innovation or measurement residual. This quantity represents the discrepancy between the measured outputs and the outputs predicted by the observer, allowing the state estimates to be corrected continuously as new measurements become available.

The optimal observer gain is obtained by solving the continuous-time Algebraic Riccati Equation

$$AP + PA^T - PC^T R_k^{-1} CP + GQ_k G^T = 0, \quad (40)$$

where P is the estimation error covariance matrix. The observer gain is subsequently computed as

$$L = PC^T R_k^{-1}. \quad (41)$$

D. Selection of Noise Covariance Matrices

The accuracy of the Kalman filter depends primarily on the choice of the process and measurement noise covariance matrices, denoted by Q_k and R_k , respectively.

Process disturbances were assumed to act on the forward velocity, vertical velocity and pitch rate states, representing the influence of unmodelled aerodynamic effects, atmospheric turbulence and modelling uncertainties. Accordingly, the disturbance distribution matrix was chosen as

$$G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (42)$$

The corresponding process noise covariance matrix was selected as

$$Q_k = \begin{bmatrix} 0.25 & 0 & 0 \\ 0 & 0.25 & 0 \\ 0 & 0 & (0.2^\circ)^2 \end{bmatrix}. \quad (43)$$

Only two outputs were assumed to be directly measurable: the forward velocity perturbation and the aircraft altitude. These measurements were corrupted by independent Gaussian sensor noise representing airspeed sensor and altimeter uncertainties. The measurement noise covariance matrix was therefore selected as

$$R_k = \begin{bmatrix} 0.25^2 & 0 \\ 0 & 0.5^2 \end{bmatrix}. \quad (44)$$

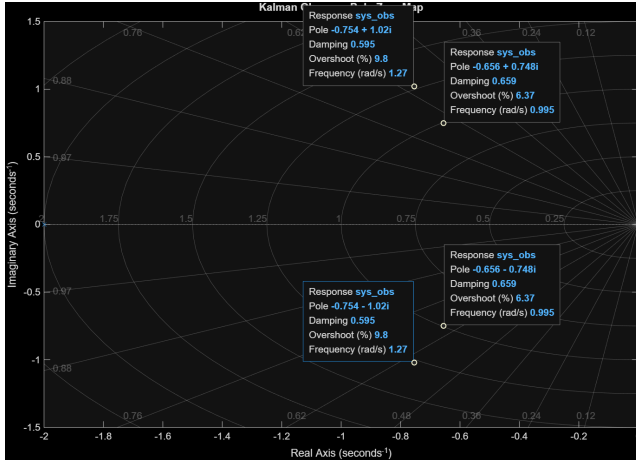


Fig. 5.1. Pole-zero map of the Kalman observer.

TABLE VI: Kalman observer eigenvalue characteristics

Mode	Eigenvalue	ζ	ω_n (rad/s)
Real Pole	-1.9987	1.0000	1.9987
Complex Pair I	$-0.7537 \pm 1.0192i$	0.5946	1.2675
Complex Pair II	$-0.6562 \pm 0.7485i$	0.6592	0.9954

The covariance matrices were refined iteratively to achieve rapid observer convergence while avoiding excessive sensitivity to sensor noise. The final values provided an effective compromise between estimation accuracy and robustness under noisy operating conditions.

E. Observer Pole Analysis

The stability and convergence characteristics of the Kalman observer were investigated by analysing its eigenvalues. Similar to the closed-loop controller, the observer poles determine the transient behaviour of the state estimates and indicate the rate at which estimation errors decay.

Figure 10 presents the pole-zero map of the Kalman observer.

The observer eigenvalues are

$$\lambda(A_{\text{obs}}) = \begin{cases} -1.9987, \\ -0.7537 \pm 1.0192i, \\ -0.6562 \pm 0.7485i. \end{cases} \quad (45)$$

The corresponding modal characteristics are summarized in Table VI.

It can be observed that all observer poles lie well within the left-half of the complex plane, confirming that the estimator is asymptotically stable. Compared with the open-loop aircraft dynamics, the observer poles are located further to the left, resulting in faster error convergence while maintaining adequate damping.

The observer therefore provides rapid and stable state estimates, allowing the estimated state vector to closely track the true aircraft states even in the presence of process disturbances and sensor measurement noise. The estimation performance

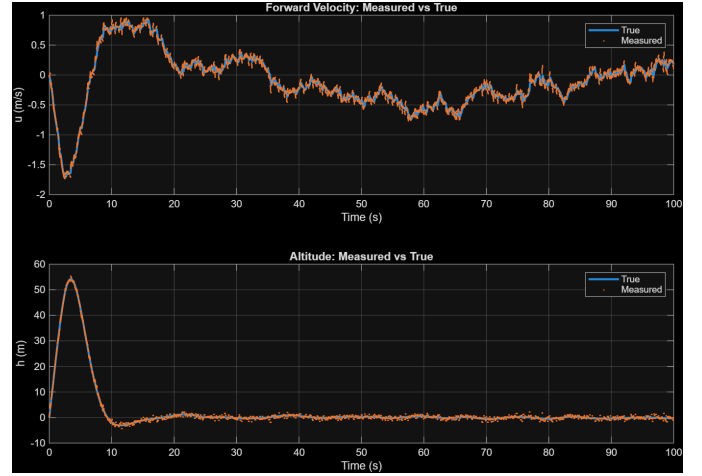


Fig. 5.2. Measured outputs available to the Kalman filter compared with the true aircraft states.

of the observer is evaluated in the following subsection using time-domain simulations

F. Observer Validation

The performance of the Kalman observer was evaluated by comparing the estimated aircraft states with the true system states under the presence of process disturbances and sensor measurement noise. Throughout the simulation, only the forward velocity perturbation and aircraft altitude were assumed to be directly measurable, while the remaining states were reconstructed entirely by the observer.

Figure 11 presents the measured sensor outputs available to the observer. As expected, the measurements closely follow the true system states while exhibiting high-frequency fluctuations introduced by the measurement noise model. These noisy measurements form the only information available to the observer for state estimation.

Using these noisy measurements together with the mathematical model of the aircraft, the Kalman filter reconstructs the remaining longitudinal states that are not directly measurable. Figure 12 illustrates the estimated vertical velocity, pitch rate and pitch angle generated by the observer.

The accuracy of the observer is evaluated by comparing the estimated states with the corresponding true aircraft states, as shown in Fig. 13. Excellent agreement is observed for all five longitudinal states, indicating that the observer rapidly converges to the true system dynamics despite the presence of disturbances and sensor noise.

To further quantify the observer performance, the estimation errors for all longitudinal states are presented in Fig. 14. Following a short transient period, the estimation errors rapidly decay towards zero and remain bounded throughout the simulation. Small residual fluctuations are present due to the stochastic process disturbances and measurement noise, but these remain sufficiently small to ensure accurate state reconstruction.

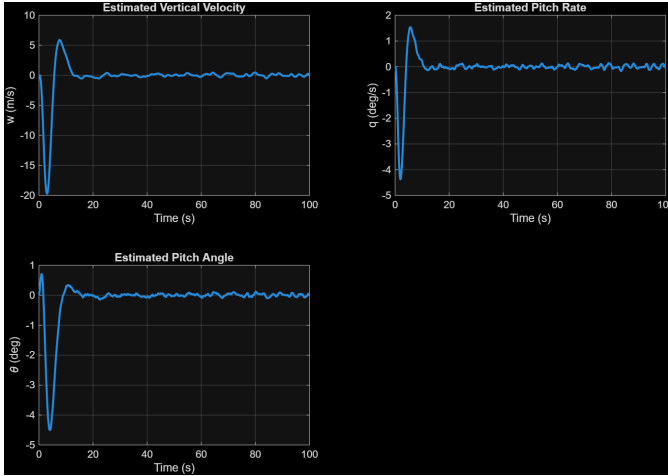


Fig. 5.3. Observer reconstruction of the non-measured longitudinal states.

The observer validation demonstrates that the Kalman filter provides accurate and reliable estimates of the complete longitudinal state vector using only noisy measurements of the forward velocity perturbation and aircraft altitude. The rapid convergence of the observer, together with the small steady-state estimation errors, confirms that the observer is suitable for integration with the LQR controller. In the following chapter, the observer and controller are combined to form the complete Linear Quadratic Gaussian (LQG) controller.

VI. LINEAR QUADRATIC GAUSSIAN CONTROL

A. LQG Architecture

Following the independent design of the Linear Quadratic Regulator and the Kalman state observer, the two components were combined to form a Linear Quadratic Gaussian (LQG) controller. The LQG controller implements optimal state-feedback using the estimated state vector provided by the Kalman filter, thereby eliminating the requirement for direct measurement of every system state.

The complete LQG architecture developed in this work is illustrated in Fig. 15.

The controller computes the control input according to

$$u = -K\hat{x}, \quad (46)$$

where $\hat{x}$ denotes the estimated state vector generated by the Kalman observer.

Unlike the LQR controller presented previously, which assumes complete knowledge of the aircraft states, the LQG controller relies solely on available sensor measurements and the observer estimates. This enables practical implementation while maintaining the advantages of optimal state-feedback control.

B. Separation Principle

A fundamental property of Linear Quadratic Gaussian control is the Separation Principle, which states that the optimal

controller and the optimal state estimator may be designed independently.

Consequently, the feedback gain matrix K is determined solely through the LQR formulation, while the observer gain matrix L is obtained independently from the Kalman filter design. Once both components have been synthesized, they may be combined directly to obtain the complete LQG controller without requiring further redesign.

The observer dynamics are governed by

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x}), \quad (47)$$

while the controller generates the control input according to

$$u = -K\hat{x}. \quad (48)$$

The Separation Principle further guarantees that the eigenvalues of the overall LQG controller consist of the union of the controller poles and the observer poles. Therefore, provided that both the LQR controller and the Kalman observer are individually stable, the resulting LQG system is also internally stable.

This property considerably simplifies the controller development process, allowing optimal control and optimal state estimation to be treated as two independent design problems before being integrated into a unified flight control system.

C. Inner-Loop Performance

The performance of the complete LQG controller was evaluated under the same disturbance and noise conditions considered during the observer design. Unlike the ideal LQR controller, the feedback law now operates entirely on estimated states generated by the Kalman filter.

Figure 16 presents the resulting longitudinal state responses.

The closed-loop responses remain well damped despite the presence of process disturbances and measurement noise. All longitudinal states rapidly converge towards their equilibrium values with only minor high-frequency fluctuations introduced by the noisy measurements. Compared with the open-loop aircraft, the controller provides substantially improved disturbance rejection while preserving smooth longitudinal motion.

D. Control Effort

The elevator and throttle commands generated by the LQG controller are shown in Fig. 17.

Both control inputs remain smooth and bounded throughout the simulation. Large control actions are observed immediately following the initial disturbance, after which the actuator commands rapidly converge towards their trim values. The absence of excessive control activity indicates that the controller successfully balances disturbance rejection with reasonable actuator effort.

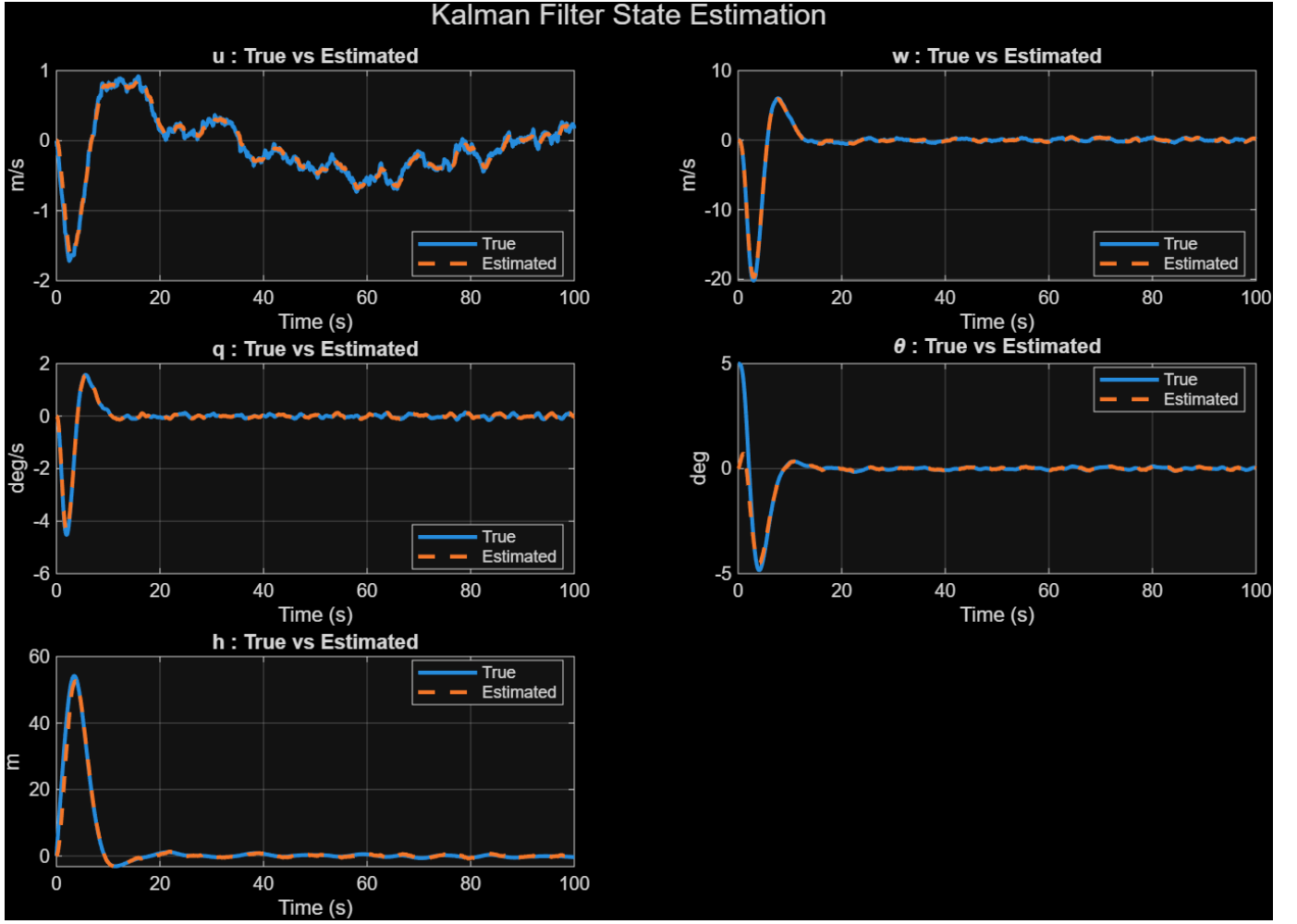


Fig. 5.4. Comparison between the true aircraft states and the corresponding Kalman filter estimates.

E. Discussion

Combining the Kalman observer with the LQR gain allows optimal state-feedback using only noisy measurements of forward velocity and altitude; despite the absence of direct measurements for the remaining states, the resulting LQG controller exhibits dynamic behaviour that closely matches the ideal full-state-feedback LQR response, confirming that the observer estimates are of practical control quality.

Having established a robust inner-loop stabilization system, the following section extends the controller to higher-level flight guidance by developing a cascaded altitude-hold autopilot, in which the LQG controller forms the inner stabilization loop while an outer-loop controller generates the required pitch commands.

VII. CASCADE ALTITUDE HOLD AUTOPILOT

A. Motivation

The Linear Quadratic Gaussian controller developed in the previous chapter provides excellent stabilization of the aircraft about its trim condition. However, it cannot independently

follow altitude commands, since its objective is disturbance rejection rather than trajectory tracking.

To enable autonomous flight, the LQG controller was incorporated into a cascade autopilot architecture. A slower outer-loop controller generates the desired pitch reference required to eliminate altitude error, while the inner-loop LQG controller rapidly tracks this command and stabilizes the aircraft dynamics.

This hierarchical structure exploits the natural separation between the slow altitude dynamics and the much faster pitch dynamics, allowing each controller to be tuned independently.

B. Cascade Control Architecture

The outer loop compares the desired altitude with the measured altitude

$$e_h = h_{\text{ref}} - h \quad (49)$$

and generates a pitch reference

$$\theta_{\text{ref}} = K_P e_h + K_I \int e_h dt. \quad (50)$$

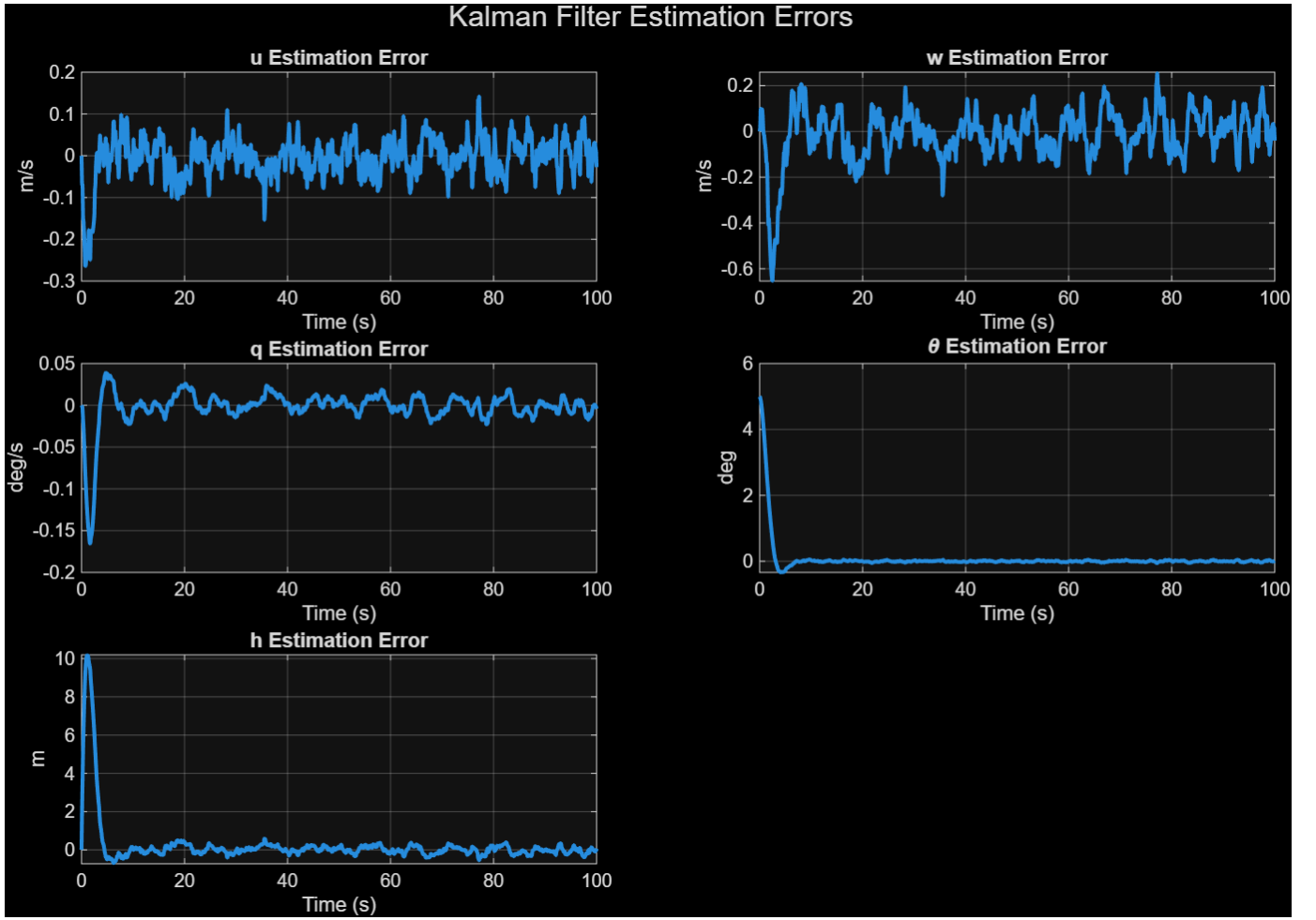


Fig. 5.5. Estimation errors of the Kalman observer for the longitudinal states.

The generated pitch command is subsequently tracked by the inner-loop LQG controller, which computes the optimal elevator and throttle inputs required to stabilize the aircraft.

C. Outer-Loop Controller Design

A PI controller was selected for altitude regulation since the derivative action provided negligible improvement while increasing the sensitivity to measurement noise.

The controller gains employed throughout this work were

$$K_P = 0.001,$$

$$K_I = 5 \times 10^{-5},$$

$$K_D = 0.$$

The proportional term determines the responsiveness of the altitude controller, while the integral action eliminates steady-state altitude error arising from disturbances and modelling inaccuracies.

D. Pitch Reference Limitation

The pitch command generated by the altitude controller was constrained within

$$-10^\circ \leq \theta_{\text{ref}} \leq 10^\circ.$$

The pitch limiter prevents unrealistic manoeuvres, reduces actuator activity and ensures that the aircraft remains within the validity range of the linearized longitudinal model.

With this limitation, the autopilot is capable of accurately tracking altitude commands of approximately 150 m while maintaining satisfactory transient behaviour. Larger altitude commands would require sustained pitch angles exceeding the imposed limit and were therefore not considered in the present study.

E. Altitude Tracking Performance

The performance of the complete autopilot was evaluated using a 100 m altitude step command. Figure 18 presents the resulting longitudinal state responses together with the corresponding Kalman estimates.

INNER-LOOP LQG CONTROLLER ARCHITECTURE

State Feedback with Kalman State Estimation

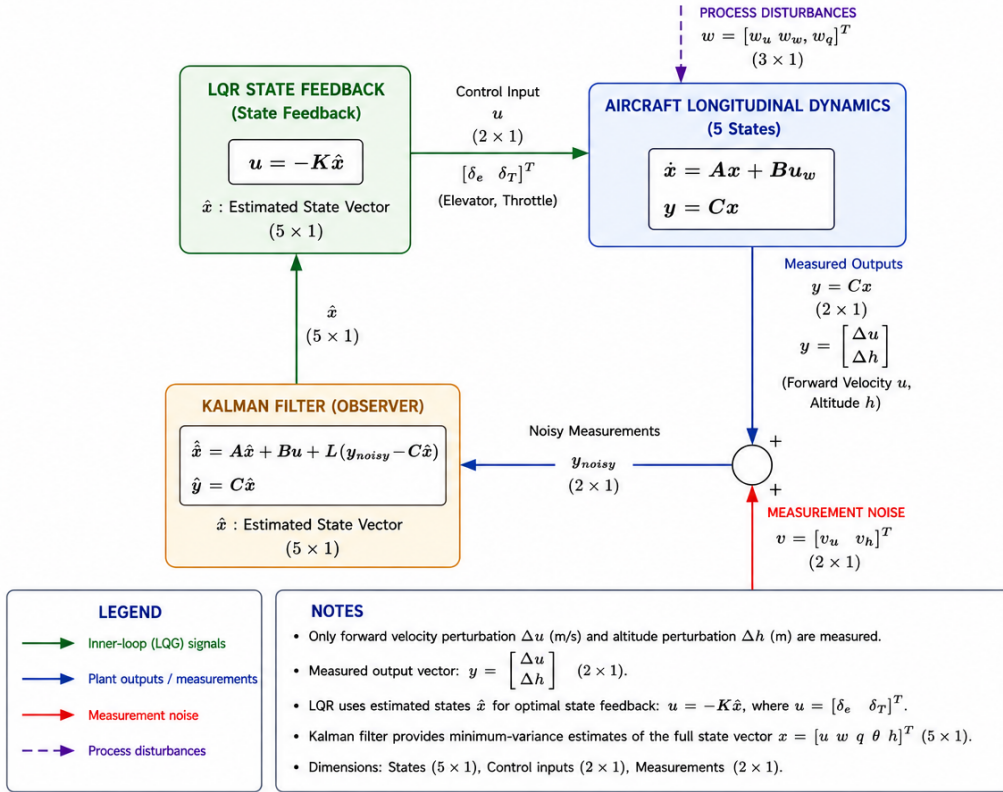


Fig. 6.1. Architecture of the Linear Quadratic Gaussian controller.

The aircraft initially pitches upward to generate the required climb, resulting in a transient increase in vertical velocity and pitch rate. As the altitude approaches the commanded value, all longitudinal states gradually return towards their equilibrium values while the altitude converges smoothly to the desired reference.

The estimated states closely coincide with the true aircraft states throughout the maneuver, demonstrating that the observer maintains a high estimation accuracy even during aggressive transient motion.

F. Control Effort

Immediately following the altitude command, the controller commands a large elevator deflection to initiate the climb. As the altitude error decreases, both the elevator and throttle commands rapidly converge towards their trim values while remaining smooth and well within practical operating limits.

G. Observer Performance During Tracking

To verify that altitude tracking does not degrade observer performance, the estimation errors for all longitudinal states are presented in Fig. 19.

The estimation errors remain small throughout the maneuver despite the large transient response associated with the climb.

Following the initial transient, all errors remain bounded close to zero, confirming that the Kalman observer continues to provide reliable state estimates during closed-loop autopilot operation.

H. Discussion

The proposed cascade autopilot successfully combines optimal state feedback, state estimation and outer-loop guidance into a unified flight-control system, accurately tracking altitude commands while maintaining stable longitudinal dynamics and bounded actuator demands. The pitch command limiter preserves the validity of the linearized model, and the Kalman observer maintains high estimation accuracy throughout the manoeuvre, together yielding a practical and robust solution for longitudinal altitude-hold control.

VIII. EFFECT OF ACTUATOR DYNAMICS

A. Motivation

In the previous sections, the control inputs generated by the LQG controller were applied directly to the aircraft model, implicitly assuming ideal actuators with infinite bandwidth. In practical flight control systems, however, the elevator and throttle actuators possess finite response times and cannot instantaneously achieve the commanded deflections.

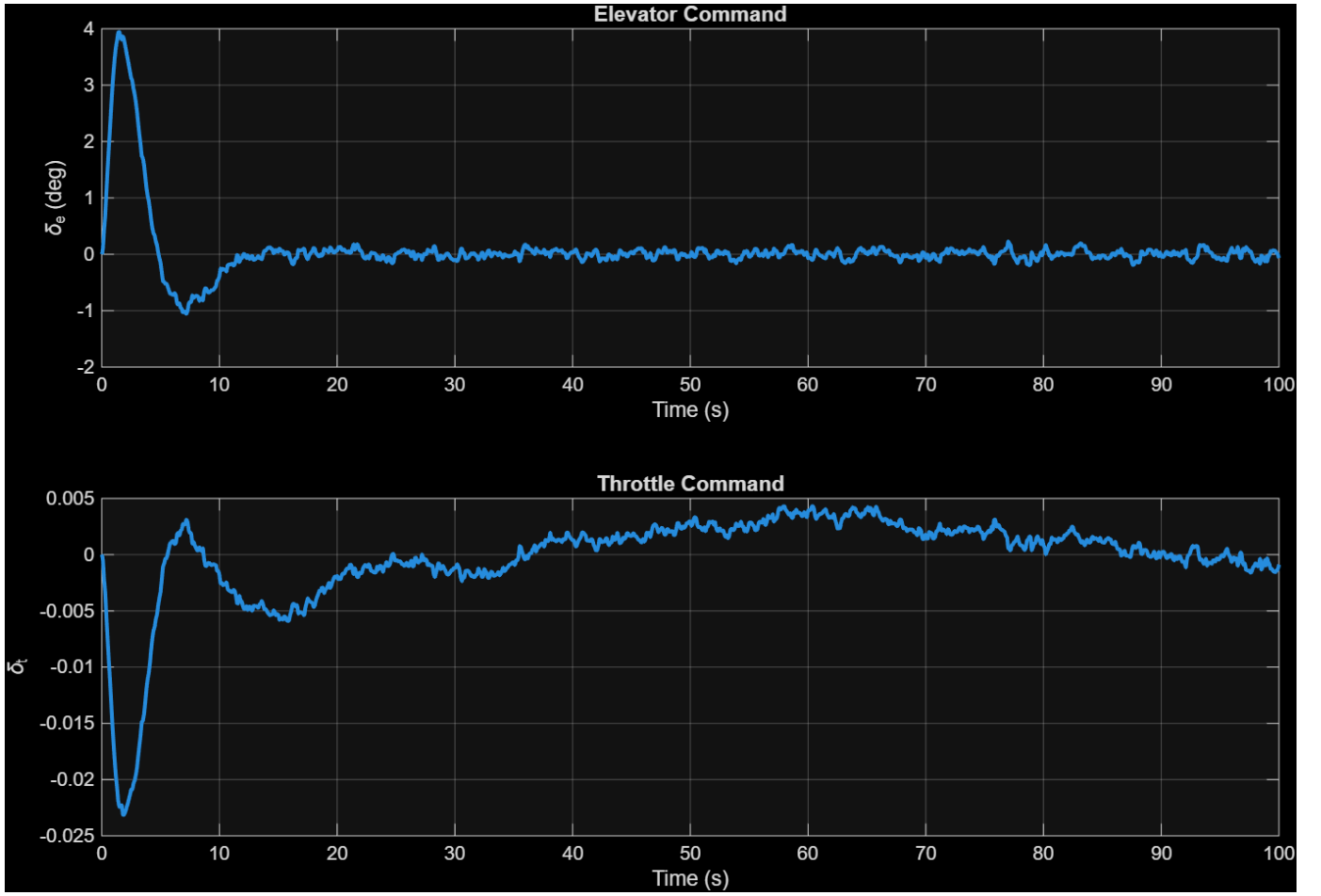


Fig. 6.2. Closed-loop response of the LQG-controlled longitudinal aircraft.

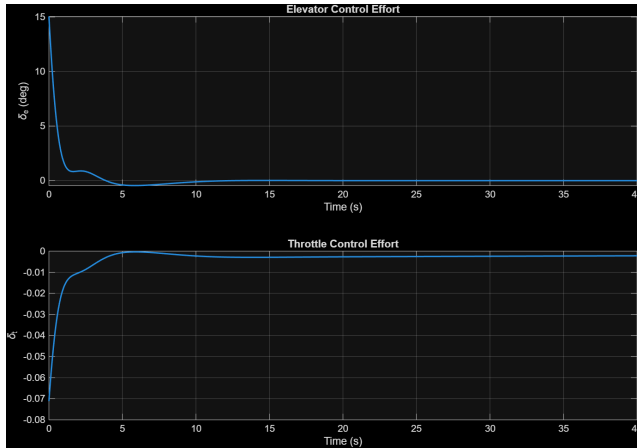


Fig. 6.3. Elevator and throttle commands generated by the LQG controller.

To obtain a more realistic representation of the aircraft control system, first-order actuator dynamics were incorporated between the controller outputs and the aircraft model. The actuator model is represented as

$$G_a(s) = \frac{1}{\tau s + 1}, \quad (51)$$

where τ denotes the actuator time constant.

Both the elevator and throttle actuators were assigned identical time constants,

$$\tau_e = \tau_T = 0.5 \text{ s},$$

which provides a realistic approximation of the finite bandwidth of conventional aircraft control actuators.

B. Effect on Control Inputs

The control inputs generated by the autopilot before and after the addition of actuator dynamics are shown in Fig. 20.

The actuator dynamics smooth the controller commands by attenuating the high-frequency components present in the ideal control signals. The initial elevator demand is no longer applied instantaneously but is instead spread over a short duration, resulting in a more gradual aircraft response.

Similarly, the throttle command becomes smoother and exhibits reduced high-frequency oscillations while preserving the overall control trend. Such behaviour is desirable in practice

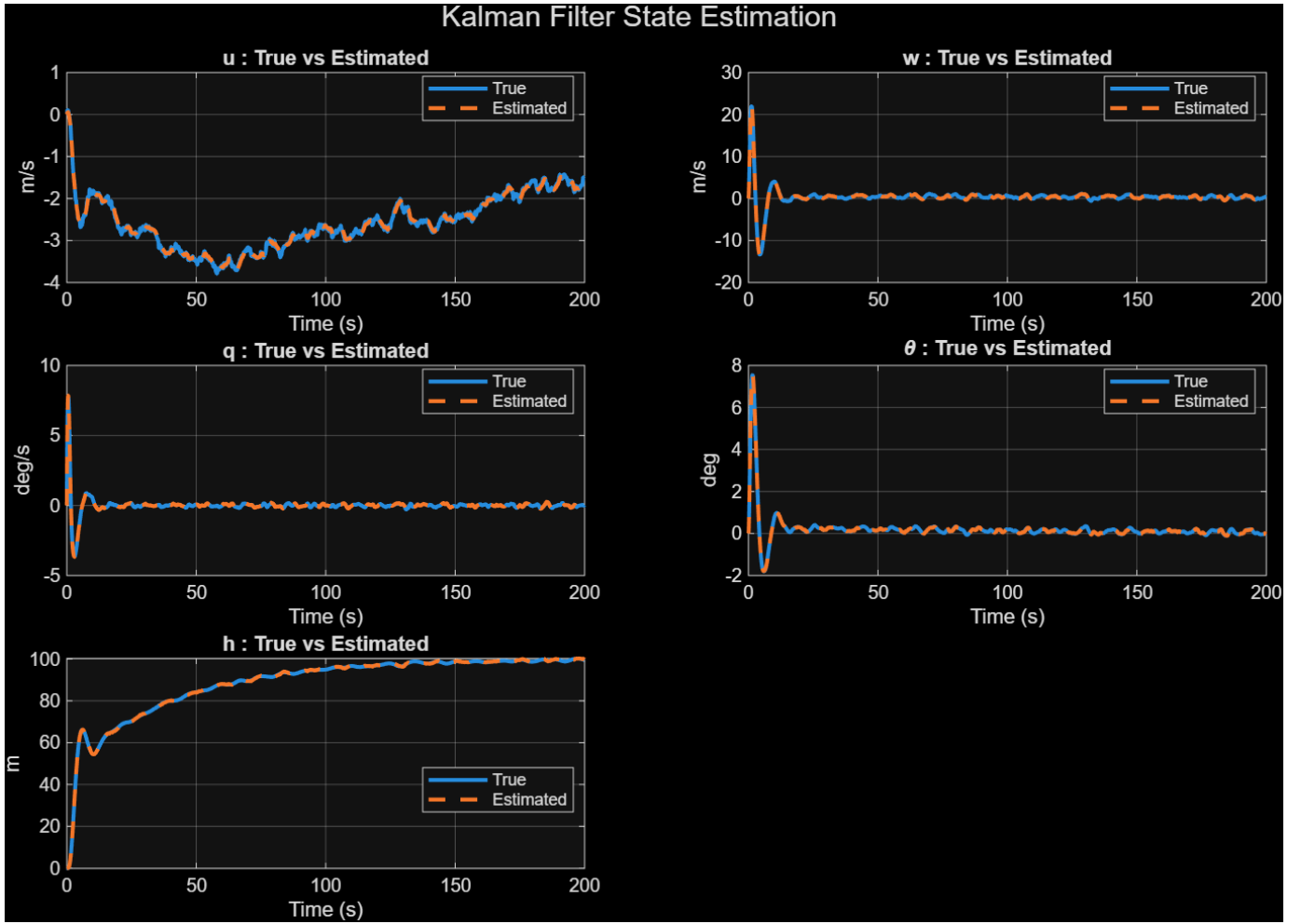


Fig. 7.1. Autopilot response to a 100 m altitude command.

since it reduces actuator wear, limits aggressive control activity and better reflects the behaviour of real flight control hardware.

C. Effect on Aircraft Response

The measured aircraft response after incorporating actuator dynamics is shown in Fig. 21.

The introduction of actuator dynamics produces a modest reduction in the closed-loop bandwidth, causing the altitude response to become slightly slower than the ideal-actuator case. The aircraft therefore requires a longer time to reach the commanded altitude, as expected from the finite response speed of the actuators.

Despite this slight reduction in responsiveness, the system remains well-damped and continues to accurately track the commanded altitude without introducing oscillatory behaviour or significant steady-state error.

D. Observer Performance

The Kalman observer continues to estimate the complete aircraft state using only the noisy measurements of forward velocity and altitude. Figure 22 presents the estimation errors obtained after the inclusion of actuator dynamics.

The estimation errors remain bounded throughout the simulation and quickly converge following the initial transient. The inclusion of actuator dynamics therefore has negligible influence on observer performance, indicating that the Kalman filter remains capable of accurately reconstructing the complete aircraft state despite the additional actuator lag.

E. Discussion

Incorporating actuator dynamics yields a more realistic closed-loop simulation: as expected, the aircraft exhibits a slightly slower transient response and smoother elevator and throttle commands, since the finite actuator bandwidth attenuates high-frequency control activity. Closed-loop stability, estimation accuracy and altitude tracking are all preserved, confirming that the inclusion of actuator dynamics improves physical realism without compromising overall controller performance.

IX. CONTROLLER PERFORMANCE EVALUATION

A. Performance Metrics

To quantitatively assess the performance of the proposed autopilot, several performance metrics were extracted directly

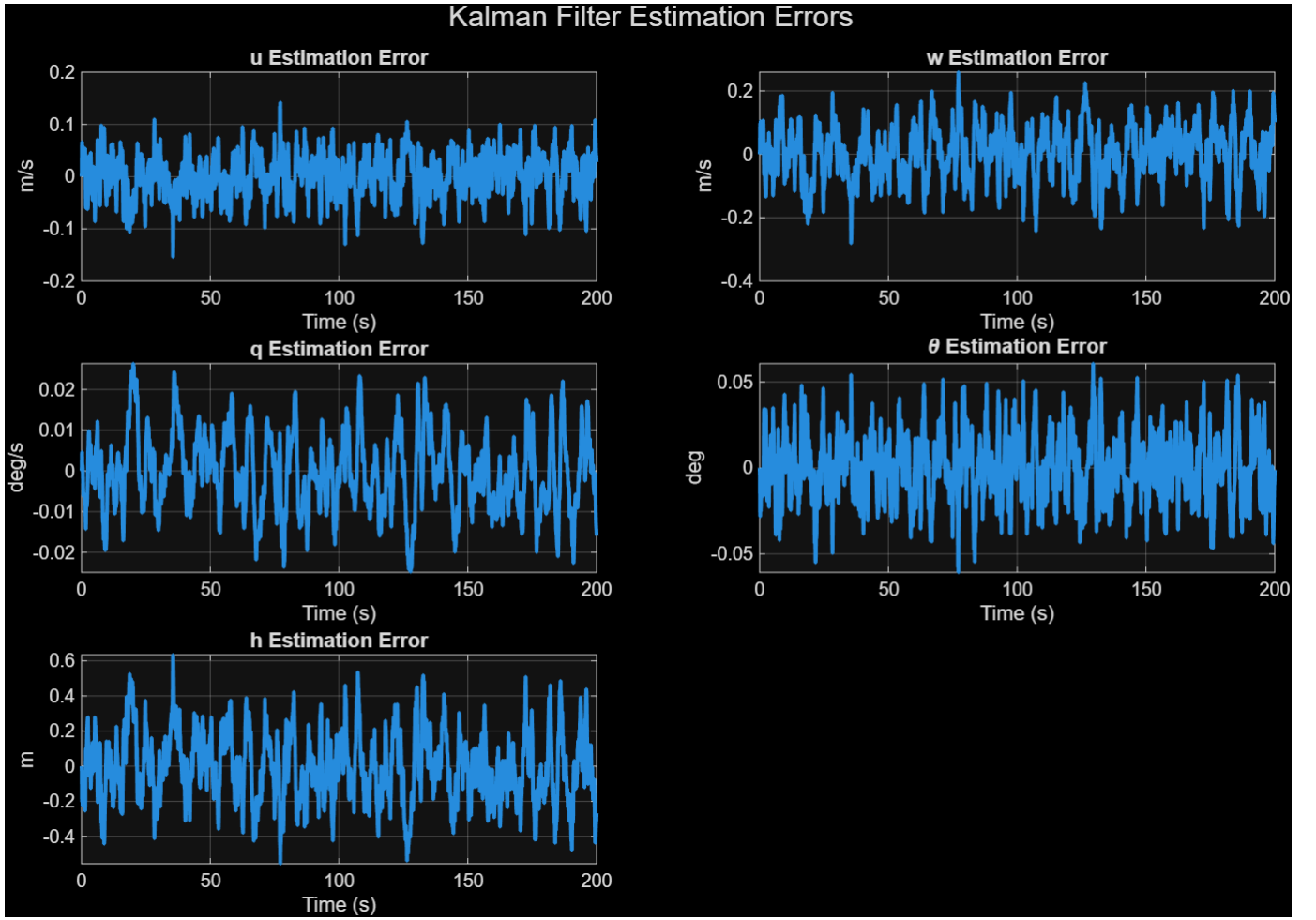


Fig. 7.2. State estimation errors during altitude tracking.

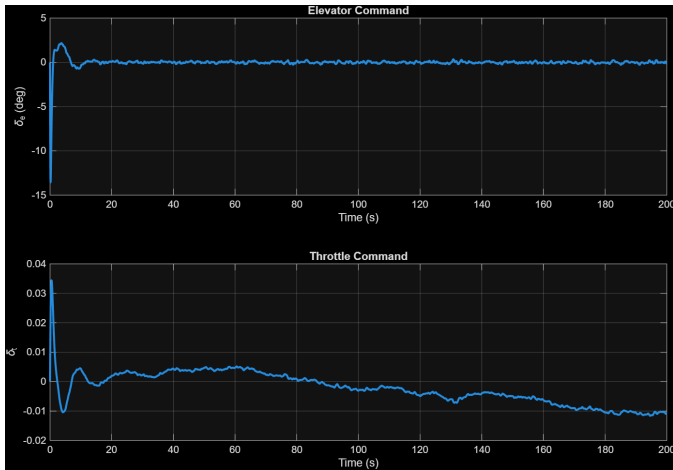


Fig. 8.1. Elevator and throttle commands with actuator dynamics.

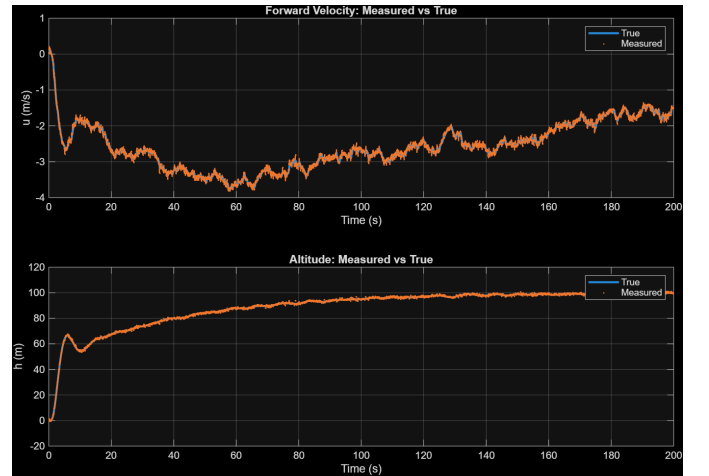


Fig. 8.2. Measured aircraft response with actuator dynamics.

from the simulation results. These include the maximum state deviations, control effort, root-mean-square (RMS) values and altitude tracking characteristics.

Table VII summarizes the principal performance indices obtained during the altitude tracking simulation.

The proposed controller achieves accurate altitude tracking with a maximum overshoot of only 0.235%, indicating ex-

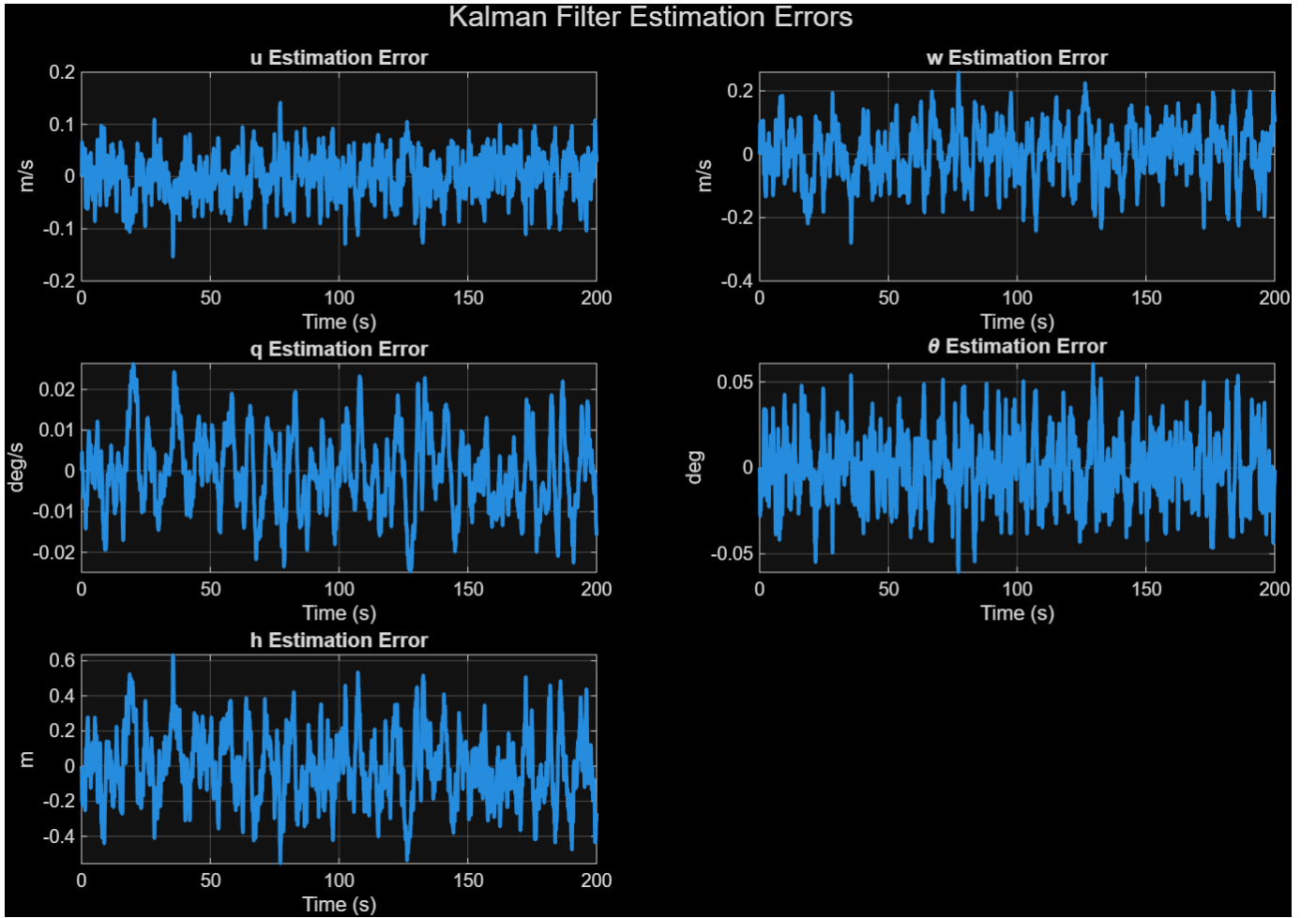


Fig. 8.3. State estimation errors with actuator dynamics.

TABLE VII: Summary of Controller Performance

Performance Metric	Value
Maximum Pitch Angle	7.90°
Maximum Pitch Rate	8.58°/s
Maximum Elevator Deflection	13.59°
Maximum Throttle Perturbation	0.0346
Peak Altitude Overshoot	0.235 m
Percentage Overshoot	0.235%

cellent transient performance. Throughout the maneuver, the commanded pitch angle remains below the imposed $\pm 10^\circ$ saturation limit, demonstrating that the outer-loop controller generates feasible reference commands while maintaining smooth aircraft motion.

Similarly, the elevator deflection remains well within practical operating limits, while the throttle perturbations are comparatively small. This indicates that the controller successfully balances tracking performance with reasonable actuator effort.

B. Overall Assessment

The simulation results presented throughout this work demonstrate the successful integration of optimal control, state

estimation and cascade guidance into a unified longitudinal autopilot.

Beginning from the open-loop aircraft dynamics, successive application of the LQR controller, Kalman observer and outer-loop altitude controller progressively improves the overall system performance. The LQR controller provides rapid disturbance rejection, the Kalman filter accurately reconstructs the complete state vector using only noisy measurements, and the cascade altitude controller enables precise tracking of commanded altitude while maintaining stable longitudinal motion.

The resulting controller exhibits smooth transient behaviour, bounded control inputs and excellent robustness in the presence of process and measurement noise, demonstrating the suitability of the proposed architecture for practical longitudinal flight control.

X. MODEL LIMITATIONS AND PRACTICAL CONSIDERATIONS

While the proposed autopilot achieves accurate altitude tracking and robust disturbance rejection within the simulation environment developed in this work, several modelling

and implementation assumptions bound the region in which these results are directly applicable to a real aircraft. These limitations are discussed below, together with experimental evidence of the practical altitude-command ceiling imposed by the linearized design.

A. Validity of the Linearization

The aircraft model used throughout this work is linearized about a single trim point, NASA Flight Condition 7 (Table I), and is therefore only accurate for small perturbations about this condition. The stability and control derivatives in Tables II and III, and consequently the matrices A and B in (19)–(20), are constants that no longer represent the true aerodynamics once the aircraft departs significantly from trim, for example following large-amplitude manoeuvres or transitions between substantially different Mach numbers or altitudes. A practical flight control system operating across a wide flight envelope would require either gain scheduling across a family of trim conditions or a fully nonlinear aircraft model with a nonlinear or adaptive controller, neither of which is considered here.

B. Altitude Command Limitation

The pitch reference generated by the outer-loop PI controller is saturated to $\pm 10^\circ$ (Section VII.D) specifically to keep the aircraft within the small-perturbation region in which the linearized model remains valid. A direct consequence of this design choice is that the achievable altitude-hold range is limited: sustained pitch angles beyond the saturation limit would be required to reach large altitude commands, but such angles cannot be generated once the limiter is active.

This behaviour was verified experimentally by commanding altitude steps of increasing magnitude and recording the resulting true altitude response, shown in Fig. 10.1–10.3. For a moderate command of $h_{\text{ref}} = 145$ m (Fig. 10.1), the aircraft converges smoothly to the reference with a peak value of 145.03 m and negligible steady-state error. As the command is increased to $h_{\text{ref}} = 180$ m (Fig. 10.2), the pitch limiter engages during the climb and the aircraft instead settles near 151.8 m, roughly 28 m short of the commanded altitude. For a much larger command of $h_{\text{ref}} = 500$ m (Fig. 10.3), the response is essentially unchanged from the 180 m case: the aircraft climbs rapidly to a peak of 156.8 m and then plateaus, leaving a large, persistent steady-state error that the integral action of the outer-loop PI controller cannot remove while the pitch command remains saturated.

These results confirm that the approximately 150 m ceiling reported qualitatively in Section VII.D is not an artefact of a single test case: once the required climb angle exceeds the $\pm 10^\circ$ limiter, the autopilot converges to a fixed altitude offset rather than the commanded reference, regardless of how large the commanded step is made. This is a deliberate consequence of preserving model validity rather than a shortcoming of the LQR/Kalman/PI design itself; a wider operating envelope would require the gain-scheduled or nonlinear extensions discussed in Section XI.

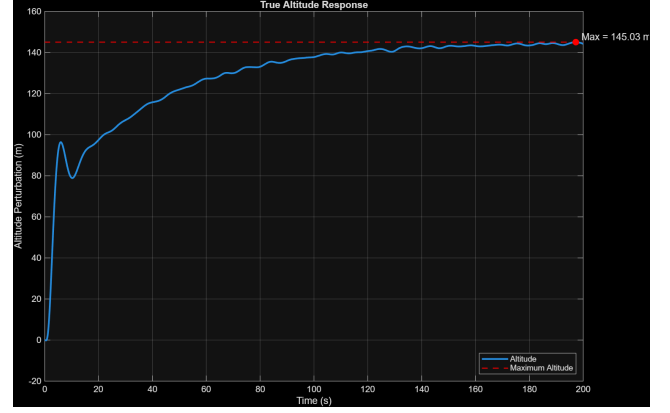


Fig. 10.1. True altitude response for a 145 m altitude command, within the pitch-limited operating range.

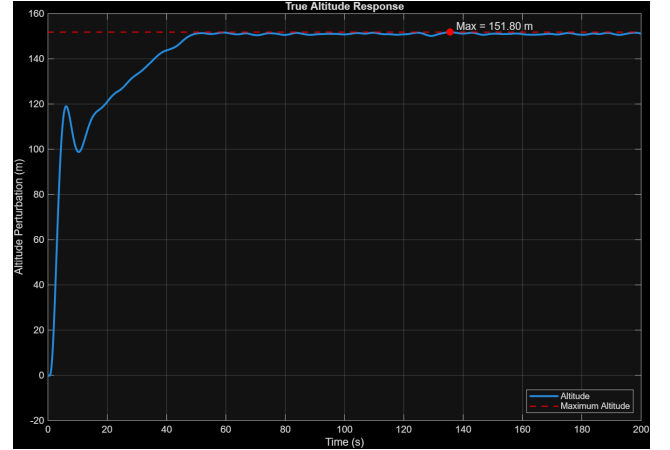


Fig. 10.2. True altitude response for a 180 m altitude command; the pitch limiter engages and the aircraft settles near 151.8 m.

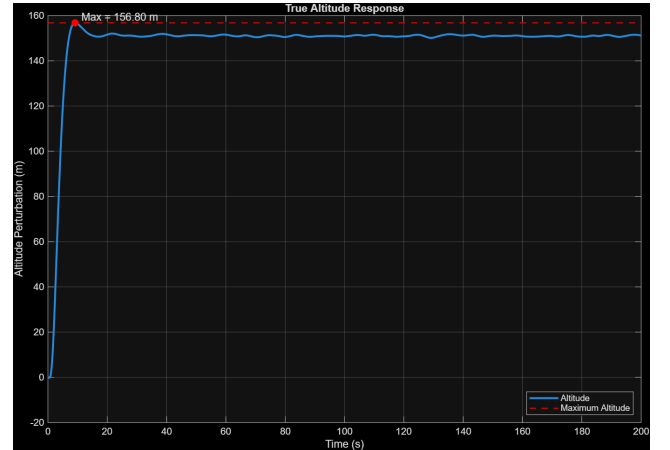


Fig. 10.3. True altitude response for a 500 m altitude command, showing saturation of the achievable altitude near 155–157 m.

C. Sensor Assumptions

The Kalman filter design assumes zero-mean, white Gaussian process and measurement noise with known, time-

invariant covariance matrices Q_k and R_k (Section V.D), and that the airspeed and altitude sensors are perfectly calibrated, unbiased, free of latency, and unaffected by quantization. Real avionics sensors exhibit bias drift, non-Gaussian noise, finite sampling rates and quantization effects that are not represented here; addressing these would typically require sensor calibration routines, bias-state augmentation, or a more general estimator such as an extended or unscented Kalman filter.

D. Actuator Assumptions

The elevator and throttle actuators are represented by the first-order transfer function $G_a(s) = 1/(\tau s + 1)$ introduced in Section VIII, which captures finite bandwidth but neglects rate limiting, hard saturation beyond the modelled deflection range, backlash, deadband and other servo nonlinearities present in physical actuation hardware. These effects would be expected to further slow the transient response and could interact unfavourably with the pitch-command limiter during aggressive manoeuvres, and should be incorporated in any hardware-in-the-loop extension of this work.

E. Environmental and Vehicle Assumptions

Finally, the simulation environment neglects several time-varying and environmental effects relevant to real flight, including wind gusts and atmospheric turbulence beyond the modelled process noise, fuel burn and the associated shift in vehicle mass and centre of gravity, and operation across multiple flight conditions with correspondingly different trim derivatives. Incorporating these effects would move the present single-trim-point design closer to a certifiable flight control system.

XI. CONCLUSIONS

This project presented the complete design and implementation of a cascaded longitudinal aircraft autopilot using modern optimal control techniques. Beginning with a linearized aircraft model obtained from the NASA Heffley and Jewell database, a five-state longitudinal state-space representation was constructed and employed for all subsequent controller design and analysis.

The open-loop aircraft dynamics were first investigated to identify the dominant longitudinal modes and evaluate the natural stability characteristics of the aircraft. An optimal Linear Quadratic Regulator (LQR) was then designed to improve the transient response and increase the damping of the short-period mode while preserving the desirable phugoid behaviour.

Since complete state measurements are generally unavailable in practical flight control systems, a Kalman filter was subsequently developed to estimate the full aircraft state vector using only noisy measurements of forward velocity and altitude. By combining the LQR controller with the Kalman observer through the Separation Principle, a Linear Quadratic Gaussian (LQG) controller capable of optimal feedback using estimated states was obtained.

The LQG controller was then integrated within a cascaded altitude-hold autopilot, where an outer-loop PI controller generated the required pitch reference while the inner-loop LQG controller performed rapid attitude stabilization. A pitch reference limiter was incorporated to maintain realistic aircraft manoeuvres and preserve the validity of the linearized aircraft model.

To improve the realism of the simulation, first-order actuator dynamics were introduced for both the elevator and throttle actuators. Although the inclusion of actuator dynamics produced a slightly slower transient response, the controller continued to exhibit smooth control inputs, accurate altitude tracking and bounded estimation errors, demonstrating its robustness to practical actuator limitations. Section X further examined the limits of this design, showing experimentally that the $\pm 10^\circ$ pitch limiter caps the achievable altitude change near 150–157 m regardless of the size of the commanded step, a direct and deliberate consequence of restricting operation to the region in which the linearized model remains valid.

Building on the limitations identified in Section X, natural extensions of this work include, in order of increasing scope:

- 1) **Gain scheduling** across a family of trim conditions spanning the aircraft's flight envelope, removing the single-point linearization constraint identified in Section X.A;
- 2) **Nonlinear aircraft dynamics**, replacing the linearized longitudinal model with the full nonlinear equations of motion to remove the small-perturbation and pitch-limiter restrictions of Section X.B;
- 3) **Full six-degree-of-freedom flight dynamics**, coupling the longitudinal and lateral-directional axes for a complete flight control study;
- 4) **Wind disturbance and turbulence modelling**, using standard gust models (e.g. Dryden or von Kármán) in place of the simplified white-noise process disturbance of Section V-B;
- 5) **Waypoint-following guidance**, extending the single-axis altitude-hold outer loop to a full three-dimensional trajectory tracking architecture; and
- 6) **Robust, adaptive, or Model Predictive Control (MPC)** formulations, which could explicitly enforce actuator and pitch constraints during optimization rather than relying on a fixed post-hoc saturation limiter.

Overall, the proposed controller successfully integrates aircraft modelling, optimal control, state estimation, guidance and actuator dynamics into a single unified longitudinal autopilot architecture, capable of stable and accurate altitude-hold operation within its validated operating envelope. Beyond its immediate simulation results, the project demonstrates that modern state-space control methods – LQR, Kalman filtering and their LQG combination – offer a systematic and practically effective alternative to manually tuned classical autopilots for coupled MIMO flight dynamics. The modular structure adopted here, in which modelling, inner-loop stabilization, outer-loop guidance and actuator realism are developed and

validated as independent, separable stages, provides a solid and extensible foundation for the more realistic flight control systems outlined above.

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NOMENCLATURE

u	Forward velocity perturbation (m/s)
w	Vertical velocity perturbation (m/s)
q	Pitch rate (rad/s)
θ	Pitch angle (rad)
h	Altitude perturbation (m)
δ_e	Elevator deflection (deg)
δ_T	Throttle command
Q	State weighting matrix
R	Control weighting matrix
L	Kalman observer gain
K	LQR state feedback gain

APPENDIX A

COMPLETE SIMULINK IMPLEMENTATION

Figure 26 presents the complete MATLAB/Simulink implementation of the proposed longitudinal altitude-hold autopilot developed throughout this work. The model integrates all major components of the control system into a unified simulation environment.

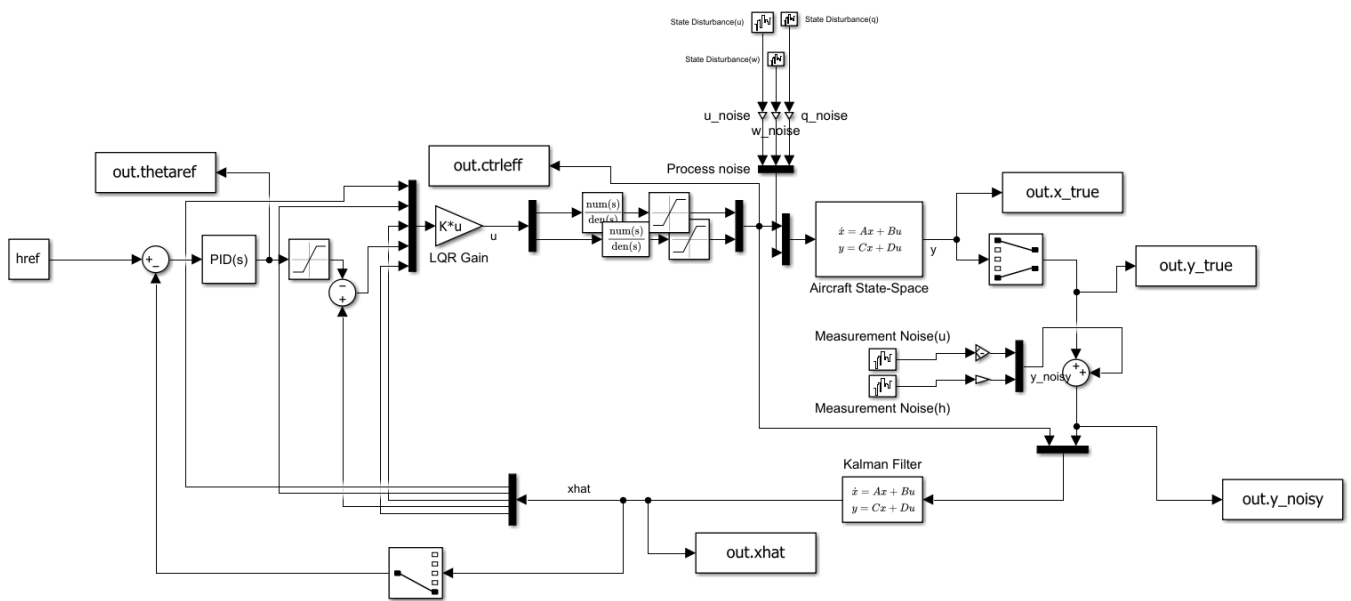
The outer-loop PI controller generates the desired pitch reference from the altitude tracking error. To ensure operation within the validity region of the linearized aircraft model, the commanded pitch angle is limited to $\pm 10^\circ$. The limited reference is compared with the estimated pitch angle provided by the Kalman observer, and the resulting error is supplied to the inner-loop controller.

The inner loop consists of the Linear Quadratic Gaussian (LQG) controller, where the optimal LQR state-feedback law operates on the estimated aircraft states reconstructed by the Kalman filter. The controller generates both elevator and throttle commands, which are then passed through first-order actuator models before being applied to the aircraft state-space model.

To evaluate controller robustness under realistic operating conditions, Gaussian process disturbances are injected into the aircraft dynamics, while Gaussian measurement noise is added to the forward velocity and altitude measurements prior to state estimation. The Kalman observer uses these noisy measurements together with the aircraft model to reconstruct

the complete longitudinal state vector, which is then fed back to the LQR controller to complete the output-feedback loop.

Signal logging blocks are incorporated throughout the model to record the true states, estimated states, control inputs, reference commands and noisy measurements used in the analyses presented throughout this report. The complete implementation therefore provides a unified framework for validating the aircraft model, controller, observer, actuator dynamics and altitude-hold autopilot developed in this work.



Complete MATLAB/Simulink implementation of the proposed cascaded longitudinal altitude-hold autopilot. The model integrates the outer-loop PI altitude controller, inner-loop LQG controller, first-order actuator dynamics, aircraft state-space model, process disturbances, measurement noise, Kalman state observer and signal logging used for performance evaluation.